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Report of the Working Group on the Long-term Sustainability of Outer Space Activities[1]

I. Introduction

1. At its sixty-second session, in 2019, the Committee on the Peaceful Uses of Outer Space adopted the Guidelines for the Long-term Sustainability of Outer Space Activities (A/74/20, annex II). At that session, the Committee encouraged States and international intergovernmental organizations to voluntarily take measures to ensure that the Guidelines were implemented to the greatest extent feasible and practicable (A/74/20, para. 163).

2. At the same session, the Committee decided to establish, under a five-year workplan, a working group under the agenda item on the long-term sustainability of outer space activities of the Scientific and Technical Subcommittee (A/74/20, para. 165).

3. The Committee also decided that the working group would agree on its own terms of reference, methods of work and dedicated workplan, and that the working group would be guided by the following framework (A/74/20, para. 167):

(a) Identifying and studying challenges to, and considering possible new guidelines for, the long-term sustainability of outer space activities. This could be done by taking into consideration existing documents, including, inter alia, documents A/AC.105/C.1/L.367 and A/AC.105/2019/CRP.16;

(b) Sharing experiences, practices and lessons learned from voluntary national implementation of the adopted Guidelines;

[1] In accordance with the multi-year workplan of the Working Group on the Long-term Sustainability of Outer Space Activities, the draft final report of the Working Group is to be finalized and presented to the Scientific and Technical Subcommittee at its sixty-third session, in 2026 (A/AC.105/1258, annex II, appendix, para. 18). At its sixty-eighth session, in 2025, the Committee on the Peaceful Uses of Outer Space noted that the Working Group had agreed that it had reached consensus on the text of its draft final report, as contained in conference room paper A/AC.105/2025/CRP.33, and that the text of that report would be formally edited and translated into the six official languages of the United Nations following the sixty-eighth session of the Committee (A/80/20, para. 107).

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(c) Raising awareness and building capacity, in particular among emerging space nations and developing countries.

4. In accordance with its terms of reference (A/AC.105/1258, annex II, appendix, para. 9), the Working Group on the Long-term Sustainability of Outer Space Activities was tasked with producing a thorough report on the long-term sustainability of outer space activities, containing the following:

(a) Information on the identification and study of challenges to the long-term sustainability of outer space activities, and corresponding recommendations, as well as possible new guidelines for the long-term sustainability of outer space activities;

(b) Information on experiences, practices and lessons learned from voluntary implementation of the adopted Guidelines and recommendations for their further practical implementation;

(c) Information on and recommendations for capacity-building and awareness-raising activities, including those related to improving international cooperation in capacity-building, taking into particular consideration the requirements of emerging space nations and developing countries;

(d) Recommendations on future activities and work on the long-term sustainability of outer space activities.

5. The Working Group followed a multi-year workplan (A/AC.105/1258, annex II, appendix, para. 18), working during the annual sessions of the Scientific and Technical Subcommittee as well as in the intersessional period, as needed, and attaching equal importance to each of the three elements of its terms of reference.

6. Many ideas and concepts considered by the Working Group were interconnected. For instance, some experiences in implementing the adopted Guidelines included or revealed a challenge or a related opportunity for capacity-building.

II. Results of the work of the Working Group

A. Information on the identification and study of challenges to the long-term sustainability of outer space activities, and corresponding recommendations, as well as possible new guidelines for the long-term sustainability of outer space activities

1. Information

7. Repeated themes or groupings of challenges, taken from the tables of compiled input on challenges provided by members of the Working Group (see the annex to the present report*), include:

(a) Large constellations, including related regulation, registration, operational contact points and need for a definition;

(b) Space situational awareness, including related information-sharing and coordination;

(c) Spacecraft manoeuvring – rules and requirements;

(d) Design and operation of small-sized space objects;

(e) Active debris removal and on-orbit operations;

* Due to spending restrictions introduced in response to the Secretariat's liquidity crisis, the annex is circulated in the language of submission only.

(f) Ensuring that all nations, including emerging spacefaring nations, can participate, in an inclusive manner, in space activities and implementation of the Guidelines;

(g) Capacity-building and international cooperation.

8. There are risks and challenges in outer space activities and difficulties in implementation of the voluntary Guidelines for the Long-term Sustainability of Outer Space Activities. Consequently, there could be a need to build upon existing guidelines or to develop new ones in areas that are not currently covered or that may be insufficiently covered, or to do so on the basis of feedback on implementation and, in areas concerning the safety and risks of space operations, and have been proposals offered in that regard. Information on those challenges can be found in documents A/AC.105/2022/CRP.11, A/AC.105/C.1/2023/CRP.4, A/AC.105/C.1/2023/CRP.17, A/AC.105/C.1/2024/CRP.32 and A/AC.105/C.1/2025/CRP.7.

2. Recommendations

9. Related recommendations include:

(a) Study further the repeated themes within the challenges referred to in paragraph 7 above, including discussion of challenges that appear in multiple guideline areas (see sections A–D of the Guidelines adopted in 2019);

(b) Give additional consideration to the following thematic areas:

(i) Space situational awareness, including related information-sharing and coordination;

(ii) Space debris mitigation: the design of space missions taking into account the Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space and the Space Debris Mitigation Guidelines of the Inter-Agency Space Debris Coordination Committee (IADC);

(iii) In-space servicing, assembly and manufacturing: the safety and sustainability of missions involving complex on-orbit operations, including satellite life extension, refuelling, the inspection of on-board anomalies, logistics and cargo servicing, space mobility, assembly in orbit, and space object de-orbiting and active debris removal;

(iv) Sustainability of cis-lunar and deep space missions;

(v) Safety considerations for human space flight;

(vi) Potential findings on dark and quiet skies and large constellations;

(vii) International cooperation and capacity-building, which is a cross-cutting theme.

B. Information on experiences, practices and lessons learned from voluntary implementation of the adopted Guidelines and recommendations for their further practical implementation

1. Information

10. Working Group members have voluntarily reported on experiences, practices and lessons learned from voluntary implementation of the adopted Guidelines. Such information can be found in documentation for the Committee, the Subcommittee and the Working Group, as well as in the Long-term Sustainability of Outer Space Activities Information Repository developed by the Office for Outer Space Affairs (see A/AC.105/1279, annex II, paras. 17–21, and https://lts.unoosa.org).

2. Recommendations

11. Related recommendations include:

(a) Continue voluntary implementation of, and share information about, the adopted Guidelines and, where appropriate, conduct outreach activities to promote implementation globally and motivate new space actors to implement the Guidelines;

(b) Use the Long-term Sustainability of Outer Space Activities Information Repository to collect information regarding voluntary implementation;

(c) Undertake an assessment of implementation mechanisms that could be relevant to addressing the long-term sustainability of outer space activities.

12. Guideline-specific recommendations include:

(a) States should be aware of the location, orbital parameters, physical characteristics and status of their in-orbit space objects, including activities undertaken by non-governmental entities, which shall be supervised or authorized by States Members, in accordance with article VI of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (see guideline B.1 (Provide updated contact information and share information on space objects and orbital events) and guideline B.2 (Improve accuracy of orbital data on space objects and enhance the practice and utility of sharing orbital information on space objects));

(b) Communication, coordination and collaboration between existing or developing space situational awareness systems in different regions of the world should be promoted, and the further harmonization of such systems explored. That could be achieved through the routine and standardized exchange of information on space objects and events in order to improve space safety and sustainability (see guideline A.5 (Enhance the practice of registering space objects), guideline B.1 (Provide updated contact information and share information on space objects and orbital events), guideline B.2 (Improve accuracy of orbital data on space objects and enhance the practice and utility of sharing orbital information on space objects) and guideline B.3 (Promote the collection, sharing and dissemination of space debris monitoring information)). In that regard, further consideration should be given to how to accomplish this objective, which could include facilitating information exchange among operators of space situational awareness systems or assessing the utility of establishing a United Nations database on space objects and events;

(c) Awareness-raising should be encouraged to sensitize all space operators to the risk of collision in an increasingly congested outer space environment and to the importance of inter-operator coordination (see guideline B.1 (Provide updated contact information and share information on space objects and orbital events) and guideline B.4 (Perform conjunction assessment during all orbital phases of controlled flight));

(d) States should cooperate with each other to avoid collisions and consider implementing navigation and anti-collision mechanisms to ensure the sustainability of current and future outer space activities (see guideline B.4 (Perform conjunction assessment during all orbital phases of controlled flight) and guideline C.4 (Raise awareness of space activities));

(e) States should consider ongoing monitoring and regular reporting to assess the effectiveness of measures for mitigating space debris (see guideline D.2 (Investigate and consider new measures to manage the space debris population in the long term)), taking into account that there is related ongoing work being undertaken, in particular under the agenda item on space debris of the Scientific and Technical Subcommittee of the Committee on the Peaceful Uses of Outer Space and by IADC;

(f) States could consider providing incentives for the development of new technologies and innovative practices to help mitigate the creation of new space debris (see guideline D.2 (Investigate and consider new measures to manage the space debris population in the long term));

(g) Public-private partnerships, where appropriate, should be encouraged to accelerate the adoption and implementation of new space debris mitigation and

remediation technologies (see guideline D.2 (Investigate and consider new measures to manage the space debris population in the long term)).

C. Information on and recommendations for capacity-building and awareness-raising activities, including those related to improving international cooperation in capacity-building, taking into particular consideration the requirements of emerging space nations and developing countries

1. Information

13. Relevant international and regional cooperation is supported through a number of organizations, initiatives and forums, including the following: African Space Agency; European Space Agency; Asia-Pacific Space Cooperation Organization; Subcommittee on Space Technology and Applications of the Association of Southeast Asian Nations; Brazil, China, Egypt, Ethiopia, India, Indonesia, Iran (Islamic Republic of), Russian Federation, South Africa and United Arab Emirates (BRICS Plus); Commonwealth of Independent States; European Union; European Organization for the Exploitation of Meteorological Satellites; IADC; International Committee on Global Navigation Satellite Systems; International Space Exploration Coordination Group; Committee on Earth Observation Satellites; Group on Earth Observations; International Organization for Standardization; Charter on Cooperation to Achieve the Coordinated Use of Space Facilities in the Event of Natural or Technological Disasters; International Space Weather Initiative; International Space Environment Service; Committee on Space Research; International Astronautical Federation; Asia-Pacific Regional Space Agency Forum and its National Space Legislation Initiative; European Cooperation for Space Standardization; World Meteorological Organization; International Telecommunication Union; International Civil Aviation Organization; Office for Outer Space Affairs; and the regional centres for space science and technology education, affiliated to the United Nations.

2. Recommendations

14. Related recommendations include:

(a) Ensure that international and regional cooperation efforts are as inclusive as possible, with particular efforts being made to include developing countries;

(b) Encourage, where appropriate, multi-stakeholder dialogue in international cooperation. This includes public, commercial and academic representatives, in addition to Governments;

(c) Encourage capacity-building and awareness-raising activities, which may take many forms, including the following: training courses, fellowships, webinars, workshops, international conferences, forums at the ministerial level, industry events, the provision of technical assistance, technology transfer on a voluntary and mutually agreed basis, academic articles, digital outreach programmes, social media efforts, podcasts and monthly question-and-answer opportunities with subject matter experts from the space sector, fostering the development of relevant and appropriate space capabilities in interested States;

(d) Continue, within available resources, support from the Office for Outer Space Affairs for capacity-building and awareness-raising activities in areas linked to the long-term sustainability of outer space activities;

(e) Encourage developed space nations to continue to share insights into and know-how related to operational and design approaches to addressing challenges related to the sustainability of outer space activities;

(f) A common approach to registering constellations to improve efficiency and completeness, based on the work of the Working Group on the Status and

Application of the Five United Nations Treaties on Outer Space of the Legal Subcommittee, could be shared by the Office for Outer Space Affairs;

(g) A glossary to facilitate the understanding and harmonization of specific terminology shared by all member States could be developed;

(h) Conduct, at the national level, outreach with industry and academia to promote the development and use of techniques and methods to improve knowledge and share that knowledge with the international community. This includes participation in conferences, workshops and exercises;

(i) Develop a list of frequently asked questions and factsheets on specific aspects of Guideline implementation/the long-term sustainability of outer space activities, as appropriate, if they could help with capacity-building among emerging and established countries.

D. Recommendations on future activities and work

15. Recommendations include:

(a) Leverage opportunities through the Scientific and Technical Subcommittee, the Legal Subcommittee and the Committee on the Peaceful Uses of Outer Space to bring experts together, under specific agenda items, to exchange best practices related to the challenges identified, as appropriate;

(b) Advance capacity-building and international cooperation related to the long-term sustainability of outer space activities;

(c) Extend the mandate of the existing Working Group on the Long-term Sustainability of Outer Space Activities and draft a new workplan that would include the decision to establish expert-oriented workstreams to focus on one or more of the agreed thematic areas of its work in order to expedite that work.

Annex*

Challenges to the long-term sustainability of outer space activities

The following tables on challenges related to the implementation of the Guidelines for the Long-term Sustainability of Outer Space Activities, challenges related to capacity-building in the framework of the Guidelines for the Long-term Sustainability of Outer Space Activities, and the identification and consideration of new challenges to the long-term sustainability of outer space activities contain substantive input and ideas from Working Group members. No consensus was reached on the input and ideas.

I. Challenges related to the implementation of the Guidelines for the Long-term Sustainability of Outer Space Activities

Section A. Policy and regulatory framework for space activities

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline A.1. Adopt, revise and amend, as necessary, national regulatory frameworks for outer space activities and **Guideline A.2. Consider a number of elements when developing, revising or amending, as necessary, national regulatory frameworks for outer space activities**		
1. Specific needs and risks associated with megaconstellations, There is a need to ensure that activities are conducted in a manner that supports the long-term sustainability of space activities and minimizes the risks of collisions and debris generation	The proposed addition to guideline A.1 is essential due to the rapid growth in megaconstellations, which require specific regulations to manage as they cause challenges in areas such as the management of radio frequency coordination, space traffic and debris mitigation. The inclusion of this paragraph ensures that States plan with long-term sustainability in mind, minimizing collision and debris-related risks. Additionally, the international harmonization of standards promotes the effective and consistent management of megaconstellations, enhancing safety and cooperation in space activities. This update strengthens the regulatory framework to address the modern demands of space operations. This challenge could be added to the Guidelines to reflect the need for specific regulations on radio frequency coordination, space traffic management and space debris mitigation due to the rapid growth in megaconstellations. The proposal was made with a view to the development of safety-related standardization without preventing the growth of the megaconstellations sector. Yes, this is a challenge that should be considered by the Working Group on the Long-term Sustainability of Outer Space Activities. Satellite constellations pose unique challenges to space sustainability at the global level, and thus the Committee on the Peaceful Uses of Outer Space and the	A new paragraph should be added to guideline A.1. The new paragraph 5 of the guideline would read as follows: “When adopting, revising or amending national regulatory frameworks for space activities, States should consider the specific needs and risks associated with megaconstellations, ensuring that activities are conducted in a manner that supports the long-term sustainability of space activities and minimizes the risks of collisions and debris generation.” The following phrase could be added at the end of the new paragraph 5: “…, along with the impacts on astronomical observations”. The following phrase could be added at the end of the new paragraph 5: “…and orbital saturation and ensures that the provision of services is in compliance with territorial rights, regulations and licensing requirements.” Proposed additional element to be added to the list in guideline A.2, paragraph 2: “Consider a high risk of collisions and space debris generation, and examine and adapt technical standards and legislation related to megaconstellations to ensure the long-term sustainability of outer space activities.”

* In the context of measures established to address the liquidity crisis in the Secretariat, the annex to the present report is made available in English only.

Section A. Policy and regulatory framework for space activities		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	Working Group, in particular, are good forums in which to address these. Given that the focus here should be on sustainability aspects related to satellite constellations, this row (i.e. this challenge) should be moved to section B (Safety of space operations), rather than section A (Policy and regulatory framework). A definition of the term "megaconstellation" is missing. It is recommended to adopt harmonized terminology in this table. There are three different terms: "megaconstellations", "large constellations" and "constellations". The preference is for the last two terms. One State does not presently have any national operator of megaconstellations. However, in the event that an operator was requesting authorization to launch and operate a large number of satellites (a definition of the term "megaconstellation" is still missing), and authorization of the launch and operation would fall under the scope of the law on space activities of 2020 of that State, that operator should undergo the authorization process and be subject to national supervision. There is no differentiation between operators controlling different size constellations. A State is committed to maintaining legal clarity and security for all types of operators. The term "megaconstellations" should be replaced with "constellations and large constellations".	A possible recommendation would be to discuss challenges associated with large constellations that fall within the mandate of the Committee on the Peaceful Uses of Outer Space in future discussions on long-term sustainability. Invite the Working Group to review the "Statement on large constellations of satellites in low Earth orbit" issued by the Inter-Agency Space Debris Coordination Committee (IADC) in relation to its consideration of the issue of large constellations. The Committee on the Peaceful Uses of Outer Space has a successful history of turning towards IADC for technical guidance on space sustainability issues; that guidance has served the Committee well in its formulation of the Space Debris Mitigation Guidelines. Additional discussion would be beneficial to understand the perceived gaps in the current guidelines with regard to the regulatory approach and safe operation/coordination of constellations ahead of confirming the need for revisions to existing guidelines or new guidelines. States are to ensure that national regulatory frameworks take into account large constellations and encourage practices (e.g. deorbiting and collision avoidance) that promote the long-term sustainability of space activities.
2. Regulation of megaconstellations	Currently, international regulations for satellite megaconstellations are inadequate and do not meet all requirements. Operating megaconstellations and providing services over the territories of States without their permission. Enacting regulations on megaconstellations shall not limit the space activities of developing countries.	A new guideline could be inserted after current guideline A.4 and could read as follows: Guideline A.4 bis. Regulation of megaconstellations 1. States should develop, revise or amend, as necessary, national regulatory frameworks for space activities specifically to address the challenges presented by megaconstellations. These frameworks should include specific requirements for the management of the radio frequency spectrum, international coordination and space debris mitigation. 2. States should encourage the adoption of international standards for the design, operation and decommissioning of megaconstellations, including coordination with the International Telecommunication Union (ITU) and other relevant entities to ensure the equitable and efficient use of orbital and spectrum resources. The phrase "and impacts on astronomical observations as well as the dark and quiet sky" could be added to paragraph 1 of the proposed new guideline.

Section A. Policy and regulatory framework for space activities		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		The phrase "and orbital saturations and ensures that the provision of services is in compliance with territorial rights, regulations and licensing requirements" could be added to paragraph 1 of the proposed new guideline. Additional discussion would be beneficial to understand the perceived gaps in the current guidelines with regard to the regulatory approach and safe operation and coordination of constellations before confirming that there is a need for revisions to existing guidelines or new guidelines.
3. Frequency coordination between non-geostationary orbit systems and geostationary orbit systems and terrestrial networks	Space resources are rare and scarce and as new non-geostationary orbit systems enter in the market, there is an increasing need for more coordination among other satellite services already in place.	Licensing framework: Regulators must foster an environment that encourages the expansion of non-geostationary orbit constellations while safeguarding the public interest. This means supporting the rollout of new technologies, ensuring a technology-neutral stance, and promoting international cooperation to prevent anti-competitive practices. The regulatory framework should be designed to stimulate innovation. To facilitate the entry of new non-geostationary orbit satellite operators, regulators are advised to adopt creative approaches tailored to the unique needs of global non-geostationary orbit constellations.
4. Regulating navigation and anti-collision	It is difficult to regulate navigation without the cooperation of all States involved in space activities. Transparency issues are evident as space object registration protocols are not properly followed by all. Opportunities for emerging space nations to participate in space activities comes at a higher risk than for earlier participants as space debris is increasing, and space debris should be considered a universal responsibility.	Suggest adding a new paragraph under the existing A.1. The new paragraph would read: "When revising or adopting regulatory frameworks, States should consider implementing navigation and anti-collision mechanisms to safeguard the sustainability of current and future outer space activities."
Guideline A.4. Ensure the equitable, rational and efficient use of the radio frequency spectrum and the various orbital regions used by satellites		
5. Mechanisms for improving equitable access to frequency spectrum and satellite orbits, including low Earth orbit (LEO) and emerging spectrum in space systems	International cooperation is at the beginning of building an equitable mechanism for low Earth orbit satellite and megaconstellation systems. Past experiences indicate that collaboration to establish an equitable mechanism has yet to succeed in establishing legal and regulatory frameworks to govern space activities and in strengthening the capacity of developing countries to use different aspects of space science technology and applications. The path towards a truly collaborative international framework is fraught with challenges as the diverse interests of nations often clash in a rapidly evolving domain. It is up to all national and international space entities to recognize that equitable access to space is a shared responsibility.	Establish a dedicated working group within the Committee on the Peaceful Uses of Outer Space to address the issue of equitable access to space-related radio frequencies and orbits for all member States. The group should develop suitable criteria to ensure that all countries can access these frequencies fairly. The proposed criteria could be based on a set of weighted factors that reflect the diverse needs and capabilities of member States. This effort should be undertaken in cooperation with ITU, ensuring that it does not prejudice the ongoing work of ITU.

Section A. Policy and regulatory framework for space activities		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline A.5. Enhance the practice of registering space objects		
6. Mechanisms for improving registration practices for large constellations, including large constellations of small sats in low-Earth orbit	The approach to registering a single satellite may be different from registering a constellation. There is no common definition of the size or number of satellites in a constellation. A common approach to registering constellations may be useful to ensure the efficient processing of registration information. Currently, there is no common definition of the size of a large constellation or the number of its satellites. Depending on the harmonized definition, it is necessary to make adjustments from the existing single space object registration approach. In addition, a unified approach taken by member States for the registration would support the efficient processing of registration information. Therefore, it should be reflected in improvements of the current registration of space objects. Yes, this is a challenge, but we see strong interlinkages to the work of the Working Group on the Status and Application of the Five United Nations Treaties on Outer Space of the Legal Subcommittee, which in 2023 considered the submission of registration information on space objects forming part of a satellite constellation and decided on a number of recommendations in that regard (see A/AC.105/1285, annex I, para. 11). If the Working Group were to study this issue, it should consider those recommendations.	Member States would be encouraged to establish a common approach to registering constellations to improve the efficiency and completeness of the database. It would be ideal to develop a glossary to facilitate the understanding and harmonization of a specific terminology shared by all Member States. The Office for Outer Space Affairs would be requested to consider what options/approaches, within existing resources, would assist their processing of the registration of satellite constellations. Proposed additional text for guideline A.5, para. 8: “States should adopt, revise and amend regulatory frameworks to improve registration practices for large constellations in accordance with the United Nations space treaties, including the Registration Convention, to which they are party.” Submissions are to be in one compiled file/batch. Constellation registration information has sometimes been submitted in batches in a single Excel file. Suggest the creation of a frequently asked questions (FAQ) document with the procedures and guidelines regarding the objects registration. The FAQ could be paired up with the glossary and be consultable through the Office for Outer Space Affairs website.
7. Improve the registration form to mention the (national) legal basis of registration	Up to now, the legal basis of registration has had to be mentioned in the accompanying verbal note, which is not clear for Member States and can lead to time-consuming correspondence with the Office for Outer Space Affairs.	Add a paragraph in the registration form.
8. Special responsibilities of launching States related to uncontrolled re-entries of space objects	In the case of several possible launching States, it must be clearly defined before the launch who is responsible. A common practice for establishing this commitment would be useful. Knowledge-sharing about insurance for space activities would be useful. If responsibility is to be addressed as a legal concept, it could be more appropriately addressed by the Legal Subcommittee. If the issue is re-entry risk analysis, it should be addressed under section B (Safety of space operations). Further clarification is needed as the challenge may have multiple components, such as the uncontrolled re-entry of large objects, which may not comply with internationally agreed approaches, and thresholds to ground risk, environmental impacts of re-entries at high-altitude due to	“Special responsibilities” needs to be further defined; and recommendations related to that are an issue separate from the challenge of “uncontrolled re-entry of space objects”. The development and transfer of design techniques, such as thermal shields and explosive bolts for the controlled disintegration of objects during re-entry, are essential. Those modifications aim to minimize the risks associated with debris that may survive uncontrolled re-entry. Establishing a platform for the updated exchange of information is crucial for timely notifications to Member States, thereby enhancing situational awareness and preparedness. There is a necessity to create capabilities for risk assessment and tracking of space objects, along with the transfer of knowledge and related technologies.

Section A. Policy and regulatory framework for space activities		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	deposition of metal particulates, de-orbit manoeuvres of constellations which move through lower orbits etc.	This initiative is aimed at designing systems that reduce the risk posed by debris during re-entry, particularly benefiting developing countries.
	Further discussion was required to understand the challenge in more detail and formulate an appropriate recommendation.	A possible best practice related to the uncontrolled re-entry of space objects would be to improve space object tracking and re-entry prediction.
	Those that are not launching States may not have specific observations to offer regarding this challenge.	
	The State agrees that this challenge is important to discuss, but additional clarification is needed.	
	Clarification is needed on this concept.	
	The challenge may relate to multiple issues, such as uncontrolled re-entry of large objects which may not comply with internationally agreed approaches and thresholds of ground risk, environmental impacts of re-entries at high-altitude due to deposition of metal particulates, de-orbit manoeuvres of constellations which move through lower orbits and different levels of space technology among States.	
9. Include additional information about objects in the national registries and registration submissions to the Office for Outer Space Affairs	Including additional information to national registries and notifying it to the Office for Outer Space Affairs is subject to the agreed forms and standardized requirements.	Considering that there is no global space situational awareness data-sharing system and that space traffic is a global issue, it might be considered useful to share the contact information of the operator assuming control of a registered object with the Office for Outer Space Affairs so that the Office can maintain a dedicated database of such data. Submit operator's contact details to the Office for Outer Space Affairs.
10. Mechanisms for improving registration practices for large constellations can relate to traditional space objects in all orbits	Consider how mechanisms for improving registration practices for large constellations can relate to traditional space objects in all orbits.	

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline B.1. Provide updated contact information and share information on space objects and orbital events and **Guideline B.2. Improve accuracy of orbital data on space objects and enhance the practice and utility of sharing orbital information on space objects**		
11. The increased need for timely and accurate	The growing importance of space situational awareness is driven by the rapid expansion of the space economy and its significant influence on the	Data-sharing. Regional cooperation.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
space situational awareness information and related data to be made available in a common/interoperable format	global economy. The challenge lies in the need to standardize and make space situational awareness information accessible in a more common, interoperable format for real-time processing. The Working Group may work on raising the awareness of timely and accurate space situational awareness information as an essential tool for identifying potential hazards and developing risk mitigation techniques. An increase in the amount of space objects and space debris requires more timely and accurate space situational awareness and data for collision avoidance. With the increased number of objects orbiting the Earth – operational or decommissioned satellites and space debris – the probability of collision and loosing assets is getting higher. Currently, data are already shared in standardized formats; so, this point is less of a challenge. The challenge lies in finding a standardized mechanism that enables information- and data-sharing and in which as many parties as possible can participate on a voluntary basis. Because the number of space objects and associated debris risk for air traffic increases at a high rate. One State agreed with the proposed inclusion of this challenge identified because if space operators share accurate information, the safety of space operations can be significantly enhanced. The notion of “timely” is not easy to operationalize. A State acknowledges this is a potential issue. However, additional discussion would be valuable to understand in detail the challenges associated with interoperability between different Member States. This would highlight the additional needs compared with the scope of the existing guidelines B.1 and B.2. To secure accurate analysis of conjunction between active and non-active objects and between two non-active objects. To improve the ephemeris calculation for astronomical image contamination for the purpose of the dark and quiet sky. Large operators are familiar with the systems available to have timely information, but this may be less the case for smaller operators. As new tools are developed, interoperability will also be fundamental. With the rapidly rising space object population and the, consequently, more frequent conjunctions, the quick and accurate assessment of the	Member States are encouraged to share information on space object and orbital events. For instance, orbital data on operational missions (with propulsion) could be uploaded multiple times per day on the United States-led free service www.space-track.org. Establishment of common platforms carrying range and angle measurements, orbital elements and conjunction data messages. Encourage all satellite operators, private and public, to subscribe to a space situational awareness system. States should develop joint strategies, as practicable, for example to increase incentives for sharing space situational awareness data and, in particular, the registration of space objects. Prevalent, widely accepted data formats can be endorsed for adoption. The first step would be to understand the current situation and available data from all member states. This would reveal the gaps and challenges. To note, continual reporting of the implementation of the existing guides will provide insight into the current practices. Upon highlighting the root cause of the challenge the approach to coordination, improving interoperability or providing additional guidance can be determined e.g. collecting guidance on approaches into a separate document, expert discussion etc. A standard protocol and format should be agreed upon and the Office for Outer Space Affairs is the best umbrella for such initiative. The Office for Outer Space Affairs can facilitate development and further improvement of recommendations on space situational awareness data technical specifications ensuring enhanced accessibility. States should encourage conjunction assessment service providers under their jurisdiction and control to prioritize disseminating timely and accurate orbital information on space objects by utilizing data in a common and interoperable format through standardized protocols for data-sharing. This information should be shared with the Office for Outer Space Affairs to ensure that all Member States have access to these critical data. A portal with updated information and a distribution process. There is a benefit to collaboration between civilian and military national stakeholders for conjunction analysis and space situational awareness. In addition, the ability to analyse space situational awareness data from

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	space situation is essential for protecting space assets. Availability of relevant data in standardized, interoperable formats and in a timely manner greatly facilitates analysis and expedited response. This challenge is connected with the lack of a centralized platform for sharing observations from space sensors and for requesting information. This challenge is connected with the absence of a dedicated communication channel between space sensors for sharing tracking requests, failure information and other safety-related data. Challenges in space situational awareness include: 1. The need for data standardization to ensure compatibility across diverse systems. 2. The urgency of timely data collection and dissemination. 3. The resource limitations faced by developing countries, in addition to inadequate data-sharing. 4. Shortage of technical expertise required to analyse and interpret the data effectively. 5. Lack of equitable access to space situational awareness data, particularly by developing countries.	multiple sources, independently verify and collaborate with international partners is important. This should be addressed through continued reporting of implementation practices to find common or agreed approaches to both the format, type of information to be shared and mechanism to ensure timely interchange. Specific guidance could be released on the implementation of certain guidelines and through targeted workshops. The Consultative Committee for Space Data Systems recommended orbit ephemeris message (OEM), orbit parameter message (OPM), conjunction data message (CDM) is followed by most agencies.
12. Improvements to space situational awareness and connected information-sharing, for conjunction-free launches and the safety of human spaceflight missions	The Working Group may consider the establishment of a common platform for information-sharing and capacity-building for emerging space countries. Space situational awareness is key in ensuring safe spaceflight for space systems carrying humans to existing space stations, as well as in consideration of the potential for commercial low Earth orbit destinations in the future and human deep space exploration missions. The increase in space debris also increases the risk for current and future human spaceflight missions. Not agreed as a challenge because of the limited number of launching entities. Enhanced space situational awareness and debris mitigation capabilities support the ongoing increase in space activity and address the related safety and sustainability challenges. Due to the increasing complexity of the orbital environment, space-based assets are increasingly at risk from collisions with other operational spacecraft or debris. At the same time, objects can re-enter and cause	A State developed and maintains its national Conjunction Risk Assessment and Mitigation Service (CRAMS) to evaluate status of orbital data and promote best practices in risk assessment. A possible recommendation would be for all spacefaring nations to have awareness of the location and status of their in-orbit space objects. A State's licence holders have to provide specific orbital parameter information and system capabilities (attitude and propulsion control systems, navigation and control capabilities, etc.) as part of the licensing process, and this requirement should be encouraged among spacefaring countries as a best practice. Ideally, non-crewed missions in the vicinity of the inhabited region should be avoided to the extent possible to avoid the build-up of the object population and to minimize potential collision risks to crewed spacecrafts. All satellites within 400+/-30 km should be manoeuvrable and trackable, and transparently share their orbital information and manoeuvre plans while transiting. Encourage discussions for global procedures on information sharing to avoid approaching and colliding through discussions in the international community.

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	damage on the ground. Monitoring and tracking such objects is needed, as well as making this information available to a wide range of stakeholders. The sharing of information and the clear designation of point of contact is the main focus here. If the increased need for timely and accurate space situational awareness information and related data to be made available in a common/interoperable format and improved mechanisms to locate the appropriate points of contactor operational communications are taken into account, this would also address the aspects of launches and human space flight. Global coordination of procedures to avoid approaching and colliding in orbit is not internationally developed as well as reflected in national practice. Additional discussion would be beneficial to understand the challenge in more detail. There is a need to confirm whether this challenge is specifically related to launch collision analysis or safety of crewed stations – throughout the full life cycle of a space mission – from risk of debris. It should be noted that with respect to launch collision analysis, the International Organization for Standardization (ISO) is gathering information on this topic which could highlight common approaches (common thresholds, objects, etc.) Special considerations must be made to safeguard astronauts against on-orbit collision risks which may arise due to non-trackable, non-manoeuvrable objects. It is understood that all Member States holding space situational awareness information must make an effort to share this information, as conjunction analysis is vital for all users of the space. In cases of manned missions, it becomes even more relevant. This also relates to the lack of a centralized platform for sharing observations from space sensors and requesting information and to the absence of a dedicated communication channel between space operators for sharing tracking requests, failure information and other safety-related data. Information is needed regarding the appropriate interpretation and utilization of collision assessments, including pre-launch assessments and pre-launch notifications containing information related to the launch schedule.	The Office for Outer Space Affairs, in collaboration with Member States, can facilitate support programs aimed to conduct new scientific missions aimed at improvement of space situational awareness. Online and on-site workshops Additional discussion would be beneficial to understand the challenge in more detail and develop an appropriate recommendation. The necessity of timely sharing and refining orbital data of space objects as a fundamental requirement for collision-free launches and safety in crewed spaceflights. The need to empower emerging countries in the development of precise launch capabilities and collision distance estimation. The necessity of empowering emerging countries (in the early stages of developing launch capabilities) to conduct collision risk assessments and analyse minimum distances. Establishing global information-sharing networks with equal and controlled access (while maintaining confidentiality considerations). In collaboration with the United Nations, Alikhanyan National Science Laboratory developed the Space Environment Viewing and Analysis Network (SEVAN) detector for space weather monitoring. The detectors are installed in various locations worldwide. The data collected by SEVAN detectors are shared internationally, contributing to a global understanding of space weather phenomena. Licence holders in one State have to provide specific orbital parameter information and system capabilities (attitude and propulsion control systems, navigation and control capabilities, etc.) as part of the licensing process, and this requirement should be encouraged among spacefaring countries as a best practice. The European Union Space Surveillance and Tracking provides a regional network of ground-based and space-based sensors capable of surveying and tracking space objects, together with processing capabilities aimed at providing data, information and services on space objects that orbit around the Earth (https://www.eusst.eu). Launch information is shared, for example, on the website of The Hague Code of Conduct against Ballistic Missile Proliferation. A Government is working on the formulation of guidelines for the prevention of collisions with satellites and other objects.

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		At end-of-life, satellites and orbital stages of launch vehicles operating above 370 km should be moved to an orbit well below the inhabited orbital regime. While performing conjunction assessment with assets belonging to human space missions, additional safety margins should be imposed.
13. Improved mechanisms to locate the appropriate points of contact for operational communications	Finding the right points of contact is a challenge that could be addressed to enable clear communication to and among satellite operators and operation centres. Establishing communication channels and mechanisms for improving communication is a critical challenge for future traffic management in space. National legislation and regulations regarding the publication or sharing of personal information may be a relevant consideration in discussions about this challenge. Secure proper interfaces between satellites operators and other users. Mechanisms to locate the appropriate points of contact for operational communications should be implemented regardless of the size of satellites. Points of contact are not shared for many non-manoeuvring objects, the main argument being that the objects cannot move and therefore points of contact do not need to be provided. However, any information from operators may be useful to enhance position knowledge and precise collision risk (e.g. data from global navigation satellite system (GNSS) sensors, attitude of the objects, etc.). Smaller States, depending on their internal organization and distribution of responsibilities, might not have an authority responsible for the activities described in guideline B.1. Guideline B.1 requires a very specific competency and capability. Operators in this State are registered with the surveillance and tracking services (United States and Europe) and handle the incoming incident reports on their own using appropriate tools. The space agency, as a public authority, does not play an active role in these matters. It is important to involve private operators in the implementation of the relevant guideline. Contact information of operators is sometimes not readily available or is completely absent, while coordination and exchange of information among operators is essential for spaceflight safety. Such exchanges are all the more important given the phenomenal rise in the number of active satellites, the majority of which belong to large constellations.	States and intergovernmental organizations should consider mechanisms to allow the sharing of contact details of satellite operators and operation centres. To create a platform where registered users (e.g. the object's owner) can find the contact information for other registered users (e.g. satellite operators). More communication between space situational awareness providers. A potential solution would be to use the space surveillance and tracking service provider tools for maintaining a database with the operational contacts. Such a database shall be available to different space surveillance and tracking service providers. The Committee shall reflect the role that the Office for Outer Space Affairs could play in collecting the operational contacts information during the satellite registration process. The United Nations Register of Objects Launched into Outer Space could expand the comprehensiveness of the database's information to include operator and/owner information. In one State, operators provide their contact information on space track. This has been proven effective and relevant to allow quick access to relevant party in case of event. There is a need to better understand registration modernization and available data, for all Member States, in order to develop an appropriate recommendation. In one State, licence holders are required to provide contact information of the individuals involved in operation of the satellite communication and remote sensing space system. This practice could be adopted more broadly, or operator information could be shared in a more centralized way. In case contact points for none or only some operators have been added, a national contact point should also be added to cover all remaining operators from that country. The national contact point should be encouraged to maintain an updated national list of operator contact points.

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	In its current state, the United Nations Register of Objects Launched into Outer Space database may lack practical (real-time) value for space operators. This is not agreement that this constitutes a challenge because each country can designate a point or points of contact.	
14. Improved mode of inter-operator coordination	Improved inter-operator coordination could not only enhance awareness of operations and reduce conjunction risk but could also serve as a transparency- and confidence-building measure. A State agrees that there is a need for an improved mode of inter-operator coordination, because the establishment of communication channels and mechanisms to improve communication is a key challenge for future traffic management. A State supports the designation of an accessible list of points of contact for each operator. Sixty per cent of collision risks involve two active objects. Acknowledging the increased risks of collisions in low Earth orbit between active objects, there is a crucial need for coordination between operators. Currently most operators rely on email-based communication, which often involves considerable delay due to time zone differences. This challenge is connected to the lack of a centralized platform for sharing observations from space sensors and requesting information. This challenge is connected to the absence of a dedicated communication channel between space sensors for sharing tracking requests, failure information and other safety-related data.	The Committee shall consider implementing common formats for data-sharing and, subsequently, developing standards. Encourage all operators to subscribe to a collision avoidance service. A common platform where the relevant data are accessible, with proper credentials and in a secured manner, can be envisaged and implemented. States should encourage operators to provide contact information both nationally and to other relevant space situational awareness service providers. Establishing contact points with the operators that own assets in the neighbourhood and are likely to be involved in potential close approaches can prove to be helpful.
15. Making available the contact information of small-satellite operators for coordination and data exchange to mitigate collision risk	Most of such satellites are launched for technological demonstration and are short-lived. The operators often forgo sharing the contact information of the teams responsible for the collision risk management. As a result, coordination to resolve any collision risk becomes difficult in the case of a close approach with such small satellites. Most emerging space countries rely on small satellites. More attention should be given to small satellites because of their rapidly increasing number. Guidelines B.1 and B.2 already provide the necessary text for the exchange of contact information between States on a voluntary basis. In our view, there is no need to differentiate between small-satellite missions and satellite missions more broadly. All satellite missions can cause	Better awareness-raising to sensitize all space actors about the collision risks in increasingly congested outer space and the importance of inter-operator coordination. Member States should ensure that all operators, including those of small satellites, provide their contact information and maintain a compilation in a centralized repository. There should be a separate identified contact point for collision risk mitigation-related data exchanges. An updated guideline on space object registration can include necessary indicative text suggesting that national focal points on space object registration collect and submit the contact information of satellite operators as well.

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	collisions. States should be able to provide the contact information about small-satellite operators within its territories. Rather than talking about small satellites, one should consider, instead, non-manoeuvring spacecraft, as the size is not a relevant distinction. Not agreed as a challenge as this should be done at country level through country guidelines. National legislation and regulation regarding the publication or sharing of personal information may be a relevant consideration in discussions on this challenge. Smaller States, depending on their internal organization and distribution of responsibilities, might not have an authority responsible for the activities described in guideline B.1. Guideline B.1 requires a very specific competency and capability. Operators from this State are registered with the space surveillance and tracking services (United States, Europe) and handle the incoming incident reports on their own, using appropriate tools. The space agency, as a public authority, does not play an active role in these matters. It is important to involve private operators in the implementation of the relevant guideline. This challenge interconnects with the lack of a centralized platform for sharing observations from space sensors and for requesting information and the absence of a dedicated communication channel between space operators for tracking requests, failure information and other safety-related data. Discussions should go beyond small satellites.	
16. A standardized method of risk assessment and a common protocol for collision avoidance	This is an important issue. As it relates to technical details, they are already addressed by international standardization bodies such as ISO (see, for example, ISO/DIS 23705 (Space systems – Identifying, evaluating, and avoiding collisions between orbiting objects), ISO 19389:2014 (Space data and information transfer systems – Conjunction data message) and ISO 21740:2025 (Space systems – Launch window estimation and collision avoidance), which the Working Group could take into consideration. The Working Group may study the common standard and adopt the methodology of implementation, through the Office for Outer Space Affairs, as a guideline. Specifying which method is actually being used can in itself be help.	A discussion between the countries to come up with such standard. For risk assessment a common scale of probability of collision will be helpful. However, we recognize the difficulties of obtaining common results owing to the different data, methods and algorithms used. All operators are likely have their own method of risk assessment. The critical part is that the source of information used by all be uniform. There are several methods in several fields. Encourage space surveillance and tracking service providers to publish info on what method is used for risk assessment. For risk assessment, in addition to the ISO standard, the United States National Aeronautics and Space Administration (NASA) has best practices, which may be beneficial for all Member States.

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	It is essential to have an international standard for effective communication. There is need for national-level discussions on procedures for collision avoidance. There are existing sources of best practices for determining risk of collision, including an ISO standard (ISO/TR 16158:2021). Additional discussion would be beneficial to understand the challenge in more detail, particularly if a "common protocol" relates to approaches to manoeuvres. The methods for risk assessment may differ, but the protocols should be put in place to know whom to contact and who should manoeuvre, or not, if conjunction is confirmed. There are already tools for this, for example, the European Space Agency Debris Risk Assessment and Mitigation Analysis (DRAMA) tool (see https://sdup.esoc.esa.int). Currently a wide range of methods to compute collision risk are used, also leading to various thresholds for the implementation of a collision avoidance manoeuvre. In some cases of conjunction between two active spacecraft, a risk can be detected by one operator and not by the other. However, it may be difficult to agree on a common approach to compute risk or on a common collision avoidance threshold. A State promotes: global coordination between space situational awareness systems that can communicate, cooperate and exchange information. Each space situational awareness system is free to retain its own methods, know-how and software, and is encouraged to share information in the event of a risky conjunction. It is even more important due to the tendency of large constellations.	Additional discussion would be beneficial to understand the challenge in more detail and develop an appropriate recommendation. For risk assessment, potential recommendations could include providing guidance or more examples of implementation with respect to existing standards or best practices. Performing detailed calculations for risk analysis using advanced modelling algorithms to predict the path of movement of space objects and calculate the probability of their collision with other objects (by using the data obtained from tracking and orbit modelling, the probability of collision between two or more space objects is calculated.) Assessing the possible effects of the collision, such as the production of new space debris or damage to satellites or space stations. This assessment helps determine the level of significance of the encounter. Common protocol for collision avoidance after risk assessment, common protocols should be implemented between all space operators for collision avoidance: 1) Early warning announcements of the early warning system: creating a system that will warn the related operators as soon as the possibility of a collision is detected. This warning should include detailed information about the time and place of possible collision. – Standard communication channels between space operators. Space operators must use common, encrypted communication channels to quickly share information about potential collisions. 2) Deciding on avoidance measures, analysing the feasibility of avoidance and examining and analysing the measures that can be taken to avoid the collision. These actions include manoeuvring one or both space objects. – Circuit change planning: If there is a need to change the circuit, a plan for this should be developed quickly. 3) Implementation and supervision of the change of execution circuit by the operators: Space operators are responsible for the implementation of the change of circuit according to the developed plan. These measures should be carried out using standard operating protocols and with the coordination of all relevant institutions. After a circuit change is implemented, its effects should be carefully monitored to ensure that the collision risk is fully reduced.

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		4) Post-incident assessment, review and analysis: After the collision avoidance process is completed, a full review should be conducted to ensure the correctness of operations and decisions. – Sharing experiences: sharing results and experiences with the international space community to improve future procedures and protocols. – Establishing and implementing international agreements to ensure that all countries and space operators follow standard procedures and collision avoidance protocols. – Supporting space laws: developing space laws that require cooperation between space agencies and guarantee access to information and data-sharing. – Holding events and workshops to share knowledge and experiences.
17. Air traffic coordination during the passage of space objects in airspace	It is necessary to coordinate closely with air traffic management in order to avoid collisions between launch vehicles and aircraft as well as to ensure a timely and effective response in emergency situations. Furthermore, pre-adjustment with authorities of the air traffic management could improve efficiency for public utilization especially when specific airspace needs to be used for the passage of space objects during a specific time. This is an important issue regarding the expected increase of re-entering space objects. Corresponding procedures are defined by national and international authorities, e.g. international civil aviation (and also maritime) organizations, which the Working Group could take under consideration. Already handled via the declaration of Notice to Air Missions (NOTAM) – notification from an official body alerting airspace users to hazards along their route, both in the air and on the ground – upon launch and risky re-entries. Air traffic notification to users – notice to air missions. It poses a safety risk to air traffic. For re-entry events related risks. Regulation concerning the re-entry of space objects falls also into the mandate of the Committee on the Peaceful Uses of Outer Space, as it does not concern air traffic coordination only – International Civil Aviation Organization (ICAO). This is an important issue regarding the expected increase of re-entering space objects. Corresponding procedures are defined by national and	There is a proposed additional guideline to follow the existing guideline B.2, which could read: “States and international intergovernmental organizations should perform appropriate coordination for air traffic management in the passage of space objects in airspace to ensure the long-term sustainability of space activities. Consider developing integrated communication protocols, airspace use notification systems, and relevant regulatory frameworks. States and international intergovernmental organizations should encourage safe and efficient operations to promote interoperability and transparency.” An interface for coordination shall be established between civil aviation authorities, air traffic services units, authorities in charge of launching activities and units for surveillance and forecast of space debris. A timely and standardized manner of exchange of information shall be observed in order for responsible entities to be able to adopt precautionary measures or react. The accuracy of trajectory data of space objects’ launching or re-entry shall be enhanced. Coordination with ICAO. Space Control Centre will work together with air traffic control department. Space and air are under Air Force responsibilities. It is suggested that this may be better addressed by ICAO and by improving linkages between ICAO and the Committee.

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	international authorities, e.g. international civil aviation (and also maritime) organizations, which the Working Group could take under consideration. No, not a challenge, as the current practices are adequate so far. There is a close relationship with the airspace control authority, as it is exercised by the Air Force itself through a Department of Airspace Control. The term “notification” is proposed instead of “coordination” for air traffic during the passage of space objects in airspace.	
18. The exchange of operational ephemeris	Operational ephemerides are crucial for meaningful assessment of conjunction risks as they are the most accurate source of orbital information for the space object and have the provision to incorporate the effects of any upcoming manoeuvre. Sometimes ephemerides may not be shared by an operator, usually those of small satellites for demonstration missions, as they may not have the means to derive ephemerides and instead rely on less-accurate mean orbital elements published in public domain as it often serves the purpose of mission operations. Some of the operators may also not be forthcoming in sharing ephemerides due to security concerns. This is an important issue, as it relates to technical details, these are already addressed by international standardization bodies like ISO (see e.g. ISO 26900:2024, “Space data and information transfer systems – Orbit data messages”), which the Working Group could take under consideration. This should be part of the standards and coordination. State agrees with the proposed challenges because if space operators share accurate information, the safety of space operations can be significantly enhanced. Additional discussion would be beneficial to understand the challenge in more detail. It is noted that the level of detail of ephemeris to be provided is due to commercial sensitivities, and therefore this challenge should possibly be re-phased to address a specific challenge and not a request for data. Connects with the lack of a centralized platform for sharing observations from space sensors and requesting information.	The operators shall use Space Data Association as baseline. For manoeuvring satellites, especially those with low-thrust engines firing over an extended duration, it is essential to share the ephemerides to avoid incorrect assessment of close approach situation by other owners/operators. A potential best practice is that space operators are able share accurate onboard ephemeris for future operations to a reputable organization for spaceflight safety, conjunction de-risk and frequency de-confliction. Several operators make their operators’ ephemerides accessible in the public domain, incorporating the effect of upcoming manoeuvres.

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	Connects with the absence of a dedicated communication channel between space sensors for sharing tracking requests, failure information and other safety-related data.	
19. Mechanisms and standards needed to implement the Guidelines that require data-sharing or consultation	A State agrees with the proposed challenges because if space operators share accurate information, the safety of space operations can be significantly enhanced. Enables efficiency of the process. Different Member States employ varying approaches to data collection, classification, and publication. In some cases, the data that needs to be shared is sensitive, which hinders effective data-sharing practices and complicates further analysis. This is an important issue as it relates to technical details, these are already addressed by international standardization bodies like ISO (see e.g. ISO 26900:2024, "Space data and information transfer systems – Orbit data messages"), which the Working Group could take under consideration. Suggest the scope of this challenge may benefit from further Working Group discussion, to understand whether there are possible overlaps with information/work arising from the Working Group on the Status and Application of the Five United Nations Treaties on Outer Space. The existing guidelines mention "appropriate communication and consultation mechanisms" and "international intergovernmental organizations", and therefore there is a need to map the gap in these organization to understand this issue further. Currently a standard mechanism for data-sharing is lacking. Most inter-operator communications continue to be email-based; data exchanges take place based on mutually agreed mechanisms, which vary from operator to operator, adding undue complexities.	A common platform for data-sharing and streamlined procedures for sharing a particular type of data should be arrived at by consensus. The mechanisms can draw inputs from the most prevalent practices. It would be ideal to map private actors' existing initiatives related to long-term sustainability and support awareness and knowledge sharing. Establishment of targeted working groups. Consultations conducted by the Office for Outer Space Affairs with Member States to establish agreed data scope and formats will help mitigate this challenge. Consider information/work arising from the Working Group on the Status and Application of the Five United Nations Treaties on Outer Space, which is undertaking an exchange of views on the implementation of article XI of the Outer Space Treaty. This includes discussion on "...possible additional tools and practices in order to enhance the exchange of information on space activities [...] such as an easily accessible and searchable repository of such information..." (A/79/20, annex II, para. 23). See European Union Space Surveillance and Tracking: www.eusst.eu. See The Hague Code of Conduct against Ballistic Missile Proliferation website: www.hcoc.at. Consultative Committee for Space Data Systems (CCSDS) standards are already used by several space agencies, and the same can be adopted.
20. Transparency and safety assurance to encourage private sector actors to implement space debris removal activities	The active removal of space debris is likely to become necessary in order to move disused space systems away from congested orbits. The constantly increasing number of satellites and other space objects in space, including as a result of megaconstellations, could otherwise jeopardize the safe and sustainable use of space in the long term. The Working Group would be an appropriate forum to discuss basic principles in the safe and sustainable conduct of space debris removal activities, both for public and private actors.	Proposed additional guideline to follow the existing guideline B.3 could read: "States and international intergovernmental organization should promote transparency and safety assurance to implement space debris removal activities including private sector actors." States and operators of space objects should consider guidelines, procedures, best practices, etc. that enable the safe and responsible use of this type of technology. For example, operators of these technologies should be incentivised to inform others transparently about the function and use of their systems.

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	The active removal of space debris technologies is currently developed by the private sector in order to move disused space systems away from congested orbits. It is important to develop and demonstrate technologies for transparent stakeholder-agreed debris remediation, including active debris removal. Guideline B.3. addresses sharing space debris-monitoring information. However, it does not include private operators in promoting transparency and safety assurance. States and international intergovernmental organization need to take action to implement their own legislation to enhance transparency on space debris removal activities. Additional discussion would be beneficial to understand the challenge in more detail. It would be useful to understand whether this relates to encouraging active debris removal, or debris mitigation in general. There is a need to understand mechanisms for "encouraging", where this would be sharing best practice on regulating private industry, or is it direct encouragement by the Committee on the Peaceful Uses of Outer Space? The challenge could be reworded as: "Encourage private sector actors to implement transparency and safety assurance when developing space debris removal activities".	Active debris removal should be carried out transparently and safely, according to procedures published by individual countries. Additional discussion would be beneficial to understand the challenge in more detail and develop an appropriate recommendation. Recommendations may include approaches to providing information on commercial in-orbit servicing and manufacturing activities that are being performed by operators licensed by a Member State, implementation examples of how novel missions are licensed, etc. The signatories of the European Space Agency's Zero Debris Charter worked out recommendations on space debris mitigation (see the "Zero Debris technical booklet"). A Government formulated "Guidelines on a License to Operate a Spacecraft Performing On-Orbit Servicing" in 2021. Now, the Commercial Debris Removal Demonstration (CRD2 mission) has been conducted in accordance with the above guidelines, under which its objectives have been made publicly available. A State is beginning to work through regulating active debris removal activities and can share best practices as this evolves.
21. Specific details of contact points for megaconstellations	This addition will ensure our guidelines are up-to-date and relevant, addressing the rapid growth of megaconstellations. By explicitly mentioning megaconstellations, we enhance coordination and response times for conjunction assessments and critical operations, ensuring direct communication channels for managing risks effectively, and also supports the long-term sustainability of outer space activities, reinforcing our commitment to proactive and comprehensive space governance. National legislation and regulations regarding the publication or sharing of personal information may be a relevant consideration in discussions on this challenge. The term "megaconstellations" should be replace with "constellations and large constellations".	There is a proposal for a modification to guideline B.1, paragraph 1: to add a simple phrase "…including specific details of contact points for megaconstellations". See: 1. States and international intergovernmental organizations should exchange, on a voluntary basis, and/or make readily available regularly updated contact information on their designated entities authorized to engage in exchanges of appropriate information on on-orbit spacecraft operations, conjunction assessments and the monitoring of objects and events in outer space, including specific details of contact points for megaconstellations, in particular those entities that are responsible for processing incoming incident reports and forecasts and adopting precautionary and response measures. This may be achieved either by providing such information to the Office for Outer Space Affairs so that the Office can make it available, within its standing mandate and existing resources, to other States and international intergovernmental organizations and/or by providing it directly to other States and international intergovernmental organizations, with the understanding that contact information for national focal points, at a minimum, will likewise be communicated to the Office.

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Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
22. Management and coordination of megaconstellations	The rapid growth of megaconstellations poses unique challenges that require dedicated regulatory attention. This guideline will help ensure that the unique risks associated with megaconstellations are effectively managed, promoting the long-term sustainability and safety of outer space activities by developing coordination procedures, implementing robust space debris mitigation measures and periodically reviewing safety protocols. A State agrees with the need for national-level discussions on management of megaconstellations. Additional discussion would be beneficial to understand the challenge in more detail. For example, clarity on the need for "coordination of megaconstellations" and whether this relates to mission definition or operation. At a high level, there may be a need for a common definition of a constellation as differentiated from a megaconstellation. There is a need to understand whether this challenge also links to the outputs that will be developed from the new agenda item on dark and quiet skies. The term "megaconstellations" should be replace with "constellations and large constellations".	There is a proposal to add a new guideline, to follow the existing guideline B.2, specifically focused on the management and coordination of megaconstellations (a renumbering the following guidelines would be needed). The text of the possible new guideline could be: "Guideline B.2 bis. Management and coordination of megaconstellations 1. Developing coordination procedures: States and international intergovernmental organizations should develop and implement specific procedures for coordinating megaconstellations, including the sharing of information on the configuration, operation and decommissioning of satellites within these constellations. These procedures must ensure that megaconstellations do not significantly increase the risk of collisions and the generation of space debris. 2. Implementing space debris mitigation measures: Operators of megaconstellations must be required to adopt robust space debris mitigation measures, including the active removal of satellites that reach the end of their operational life and ensuring that any decommissioned satellite is safely removed to minimize the presence of debris in low-Earth orbit. 3. Periodically reviewing safety protocols: Safety protocols and operational procedures for megaconstellations should be periodically reviewed and updated based on new technologies, best practices and lessons learned from previous incidents." Additional discussion would be beneficial to understand the challenge in more detail and develop appropriate recommendations.
23. Lack of a centralized platform for sharing observations from space sensors and for requesting information	The absence of a unified system for sharing space-based sensor observations and information requests among space operators leads to inefficiencies in space situational awareness, tracking and collision avoidance.	Develop and implement a centralized platform where space operators can share sensor observations and make requests for specific tracking data or information in a streamlined and secure manner. Encourage international collaboration for the creation and management of this platform, ensuring interoperability and data security among all space operators.
24. Absence of a dedicated communication channel between space operators for sharing tracking requests, failure information and other safety-related data	There is no dedicated communication infrastructure between space operators, which limits the ability to coordinate in real-time, share vital information on satellite failures and prevent potential collisions through timely tracking requests.	Establish a dedicated, secure communication channel for space operators to share operational data, tracking requests and safety-related information in real-time. Develop protocols for the communication of urgent safety data, including a registry of contacts for rapid coordination and response in the event of anomalies or safety issues.

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
25. Consider how specific details for contact points for megaconstellations and the management and coordination of megaconstellations can relate to traditional space objects in all orbits and challenges thereof	The term “megaconstellations” should be replace with “constellations and large constellations”.	
Guideline B.4. Perform conjunction assessment during all orbital phases of controlled flight		
26. The observance of spacecraft manoeuvring rules to avoid collisions (e.g. between human-rated spacecraft, robotic spacecraft and constellations)	Currently, no commonly agreed rules exist for manoeuvring satellites. Individual operators take appropriate measures to assess collision risks and mitigate them. The increase in the number of entities with related technical capabilities make it necessary to raise this point. It is necessary to discuss situation-specific guidance, including the prioritization of space object avoidance manoeuvres, and to establish basic guidelines on how satellites should perform manoeuvres and how they should operate to avoid each other when a collision risk is predicted. There are no pre-agreed “rules of the road” for collision avoidance. Collision avoidance thresholds are generally agreed, but with no specific manoeuvre rules. It is unclear whether this point is talking about defining potential priority rules between active spacecraft (for example, constellation spacecraft should move in the case of a conjunction involving any other spacecraft). Or is it about monitoring that everything has been done as it should have been done when a conjunction occurs? Clarification is needed. Already covered in the framework of the International Standards Organization and European Cooperation for Space Standardization. The lack of manoeuvring capability in many types of satellites increases the risk of collisions between space objects. Therefore, extensive research is needed on both technical innovations in this area and the design and development requirements for satellites; such research is to be conducted collaboratively among all spacefaring nations.	Create a protocol on coordinating this kind of activity. Continued reporting of implementation which highlights approaches to manoeuvring and sources of best practices for guiding approaches. In planning the manoeuvres, post-manoeuvre close approach risks should be assessed. An important point to consider is that incorporating manoeuvring capabilities in all spacecraft across various orbits imposes significant costs on the design and development process of space products. Attaining advanced and effective levels of this capability independently may be beyond the capacity of many developing countries. There is an intention to include this topic in an ISO standard on space traffic coordination. However, the only existing source is the Space Safety Coalition’s “Best practices for the sustainability of space operations”. A space agency ensures that all post-manoeuvre trajectories are clear of any potential collision risks imminently after the manoeuvre. Plans are fine-tuned to minimize such post-manoeuvre risks. Additional safety margins are applied for assessing risks of close approach with a human-rated mission. For large constellations, necessary coordination is carried out to avoid any collision. On several occasions, large constellation operators take the responsibility for manoeuvring their spacecraft for collision avoidance.

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
27. Operational transparency (e.g. notifications of manoeuvres that may result in safety issues to other operators)	Lack of transparency can lead to short-notice conjunction avoidance manoeuvres and use up satellite fuel, thus shortening mission life or leading to debris events. The increase in the number of entities with related technical capabilities makes it necessary to raise this point. A lack of transparency is one of the main reasons for the occurrence of misunderstandings between States and operators of space objects. Promoting mechanisms to improve communication and exchange can help reduce those uncertainties and thereby improve the safety of space operations. This challenge is important to discuss, but additional clarification from the proposing country is needed. The need for information exchange has already been highlighted several times. Transparency of manoeuvres may improve the accuracy of collision avoidance and risk on orbit. Notifications of manoeuvres shall be communicated in a timely manner before the execution of the manoeuvres. What happens if the other operators involved would like to manoeuvre at the same time? To create confidence in space operations. The lack of manoeuvring capability in many types of satellites increases the risk of collisions between space objects. Therefore, extensive research is needed on both technical innovations in this area and the design and development requirements for satellites, to be conducted collaboratively among all spacefaring nations.	Spacefaring nations should provide notification of manoeuvres, especially in crowded orbits. Create a protocol to coordinate this kind of activity. Encourage spacecraft operators to share, in a timely manner after the injection and with appropriate around-the-clock conjunction assessment entities (including space surveillance systems), the necessary information to mitigate collision risks, including at a bare minimum ephemeris, covariance and manoeuvre plans. States and intergovernmental organizations should consider mechanisms and incentives that promote the exchange of information between space operators on planned manoeuvres that may result in safety issues for other space operators. Furthermore, operators of space systems should also inform each other, if practicable, about the intention of their manoeuvres in order to prevent potential misunderstandings. Continued reporting of implementation examples which highlight approaches and processes to exchanging operational information. Discuss whether it is beneficial to include nominal manoeuvre plans in registration information to highlight deployment and deorbit plans. Ensure international collaboration for the creation and management of a centralized platform for sharing observations from space sensors and requesting information, ensuring interoperability and data security across all space operators. Develop protocols for the communication of urgent safety data, including a registry of contact for rapid coordination and response in event of anomalies or safety issues. An important point to consider is that incorporating manoeuvring capabilities in all spacecraft across various orbits imposes significant costs on the design and development process of space products. Attaining advanced and effective levels of this capability independently may be beyond the capacity of many developing countries. Spacefaring nations should use services that provide a 24/7 “space watch” to monitor and report on space domain awareness, in close coordination with partners, which includes threats of debris and collision, space weather and status of space mission systems. Encourage space operators to share, within three days after the injection, with relevant entities (including space surveillance systems), the information

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		necessary to mitigate collision risks, including at a bare minimum ephemeris, covariance and manoeuvre plans.
28. Requirements related to spacecraft manoeuvrability in various orbits	It is desirable to have manoeuvring capabilities while operating in highly utilized regions such as geostationary orbit and low Earth orbit and also in the vicinity of human spaceflight regions. Manoeuvring satellites is always a dangerous and tricky activity that can affect other in-orbit objects. There is a need for a global framework that would allow the States to reach an agreement on the minimum manoeuvrability capacity in different orbits. This challenge is important to discuss, but additional clarification from the proposing country is needed. A need to explore further as to whether challenge No.18 addresses concerns identified within this challenge. However, there are ongoing discussions on the expectations for the manoeuvrability of spacecraft in different orbits. For example, organizations have suggested proposals for manoeuvrability of spacecraft operating above a certain altitude, e.g. crewed stations. Additional considerations on manoeuvre performance. The lack of manoeuvring capability in many types of satellites increases the risk of collisions between space objects. Therefore, extensive research is needed on both technical innovations in this area and the design and development requirements for satellites, to be conducted collaboratively among all spacefaring nations.	Coordination of such activity is a necessity. While operating in highly populated and utilized orbital regions, manoeuvrability should be ensured for effective collision risk mitigation. States shall consider the need for manoeuvrability for collision avoidance and decommissioning above a certain altitude. An important point to consider is that incorporating manoeuvring capabilities in all spacecraft across various orbits imposes significant costs on the design and development process of space products. Attaining advanced and effective levels of this capability independently may be beyond the capacity of many developing countries. A protocol to communicate the activity ahead of time should be set.
29. Predominance of a competitive environment motivated by commercial and political views, which will prevent the formation of an interactive and cooperative approach among Member States	This challenge is a political matter. In addition, the existing Guidelines already state: "… outer space should remain an operationally stable and safe environment that is maintained for peaceful purposes and open for exploration, use and international cooperation by current and future generations, in the interest of all countries, irrespective of their degree of economic or scientific development, without discrimination of any kind and with due regard for the principle of equity..." Additional discussion would be beneficial to understand the challenge in more detail, in order to understand the intent.	There is no best practice or recommendation. The development of commercial space activities by private entities has been in part facilitating cooperation between Member States.
30. Deployment of thousands of satellites in near-Earth space in the form of large or megaconstellations,	With the large number of deployed satellites, the congestion in different orbits will make it difficult to deploy satellites.	Deployment of large or megaconstellations should be regulated. Additional discussion would be beneficial to understand the challenge in more detail and develop appropriate recommendations.

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
which can cause orbital congestion and limit the free and equal access of other Member States to the peaceful exploration and use of outer space, which is recognized as the common interest of all humanity	Community and all users should be informed about potential deployment of megaconstellations in specific region to properly adapt to upcoming situation. Currently, several orbital bands are densely populated due to the deployment of large constellations. Operating within or in the vicinity of such constellations invariably leads to frequent and numerous close approaches with constellation satellites and hence, requires additional efforts to assess the risks of a huge number of close approach alerts and extensive inter-operator coordination for collision risk mitigation. Such considerations on spaceflight safety deter a spacefaring entity from operating in the proximity of the congested orbital regimes inhabited by megaconstellations. Currently there are no rules or guidelines that limit the number of satellites that can be deployed at a given orbital altitude. The maximum number of satellites in a constellation is driven by the safety considerations within a constellation, namely the minimum physical separation required to avoid inter-satellite collision or interference. In the absence of such regulations, a particular orbital regime can be tightly packed by a constellation to the extent that it is no longer feasible for another spacefaring entity to operate in and utilize the same orbital region. The presence of multiple densely populated orbital shells, especially those in the low Earth orbital region severely constrains the availability of collision-free lift-off timings for a launch vehicle, because multiple encounters with the satellites of the constellations are inevitable during the transits. Additional discussion would be beneficial to understand the challenge in more detail. Challenge may relate to implementation of guidelines in sections A and B (i.e. appropriate regulatory frameworks, debris mitigation approach, conjunction and collision avoidance approaches, etc.) or potential new impacts such as those highlighted in the dark and quiet skies discussion. The proliferation of thousands of satellites in near-Earth space raises critical concerns regarding space operations. Attention must be paid to the implications of unequal access to orbital resources. The current landscape often favours developed countries and wealthier nations, potentially marginalizing smaller economies that are eager to engage more in space activities.	Establishing a harmonized space regulation in which all countries can equitably benefit from the vast opportunities of outer space is an essential mission for international space cooperation. Consequently, creating mechanisms that facilitate opportunities for developing countries is crucial for upholding the principle of equitable access. Such information can be provided to the community through different forms, e.g. Scientific and Technical Subcommittee meetings, conferences etc.

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline B.8. Design and operation of space objects regardless of their physical and operational characteristics		
31. Strengthen rules regarding the decommissioning of space objects	Today, most of States agree with the 25 years rule for the decommissioning of space objects. A State recognizes that if such a rule is a great progress of these last years, further actions are needed in order to ensure an always better sustainability of space activities. A possible path toward this objective is to limit the time in orbit of each space object proportionally with its operational lifetime, but to a maximum of 25 years. For example, if a factor 3 is retained: A satellite whose operations last for 4 years would have to proceed to its decommissioning within 12 years; A satellite whose operations last for 10 years would have to proceed to its decommissioning within 25 years.	Recommend limiting the presence of a space object in outer space, including protected regions, proportionally to the duration of the operation of such space object and within an adapted maximum of time. In the event of the space objects decommissioning leading to atmospheric re-entry, the residual time in orbit may not exceed: – 3 years for systems with an operational phase of less than 1 year; or – Three times the duration of the operational phase, and in no case more than 25 years.
32. Development and implementation of criteria and procedures for the preparation and conduct of space activities aimed at the active removal of space objects from orbit	This area is still in its initial stages. Thus there are no clear protocols, and the technologies associated with it is experimental. To support the industry, extending potential number of players in the domain of active debris removal and, consequently, the diversity of technologies applied. Active debris removal should be carried out in accordance with appropriate rules. For example, verification and evaluation methods for safety assurance and information-sharing on operational plans should be included in the rules. This challenge is important to discuss, but additional clarification from the proposing country is needed. Further international discussion is needed, especially from technical aspects. Additional discussion would be beneficial to understand the challenge in more detail. Specifically, a need to understand what is meant by “criteria and procedures for the preparation and conduct of space activities”. This may relate to the design of spacecraft to allow for active debris removal or other in-orbit servicing and manufacturing activities and their associated implications. As such, this challenge may be better suited to the new challenges section. To ensure space sustainability with transparency and confidence. Active removal of space debris is a highly complex and multidimensional technical process that requires consideration of various safety, security, technological and legal considerations. These considerations have not yet been systematically codified into	Creating protocols and agreements for this type of activity. Additional discussion would be beneficial to understand the challenge in more detail and develop appropriate recommendations. International protocols. The global management and regulatory process for active debris removal must be conducted with due regard for legal issues. Space debris removal should be carried out based on standardized technical procedures that are agreed upon by the members.

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Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	standardized procedures and guidelines that are widely accepted and agreed upon, and their absence is still regarded as a challenge.	
33. Safe conduct of operations for the destruction of in-orbit space objects	This type of activity will probably generate space debris that may threaten other assets. To avoid further contamination of the space environment. The Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space state that "the intentional destruction of any on-orbit spacecraft…should be avoided" and that when intentional break-ups are necessary, "they should be conducted at sufficiently low altitudes to limit the orbital lifetime of resulting fragments". In general, the State opposes the intentional destruction of spacecraft in orbit and has supported General Assembly resolution 77/41 on destructive direct-ascent anti-satellite missile testing. Destruction of on-orbit objects results in numerous fragments that jeopardize the sustainability of outer space activities. Additional discussion would be beneficial to understand the challenge in more detail. There is a potential link to security topics, which would be outside the remit of the Committee on the Peaceful Uses of Outer Space. To support space sustainability with transparency and confidence, space actors should remove from orbit, not destruct in orbit. The formulation of frameworks, standards and rules for identifying safety considerations and their implementation, as well as the establishment of a regulatory mechanism, can pose significant challenges. The issue of space debris removal involves multiple technical and legal complexities. Coordination among stakeholder countries, management of debris reduction and other unpredictable risks are among these complexities. This subject requires further discussion.	An international consensus is required as well as very strict protocols. If unavoidable, such activities should be conducted in a manner so as to not create long-lived debris. Change the challenge to remove debris and satellites that are unusable from orbit. Effective participation of countries in the implementation of this process is contingent upon the exchange of information and the transfer of up-to-date knowledge and experience to developing countries. IADC identifies ("Intentional destructions, which will generate long-lived orbital debris, should not be planned or conducted." (sect. 5.2.3)) which is referenced in guideline A.2, paragraph 2 (f), of the Guidelines for the Long-term Sustainability of Outer Space Activities, and therefore possibly this issue is covered. Conduct of such activities in lower altitude ensures the quick decay of the resultant debris as well as better chances of tracking and cataloguing them.
34. Long-term contributions and challenges of large-scale commercial space launches	With ease of access to space and growing interest in utilizing space technologies, the number of launches is steadily increasing. The cumulative effect on the upper atmosphere of the re-entering rocket stages is likely to be adverse in the long run. The defunct satellites while re-entering also have an adverse effect on the upper atmosphere during their demise, and the surviving fragments pose a significant aggregate debris casualty risk. Additional discussion would be beneficial to understand the challenge in more detail. Specifically, it is unclear whether this relates to issues such as debris or environmental impacts etc.	To the maximum extent possible, the spacecraft or upper stages of rockets ending their lives in low-Earth orbit should be made to re-enter the atmosphere in a controlled manner to limit the ground casualty risks Usage of environmentally friendly green propellants should be promoted. The necessity of devising solutions to reduce pollution in the upper atmospheric layers caused by the persistence of various fuels and combustion products resulting from numerous space launches.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	This challenge is important to discuss, but additional clarification from the proposing country is needed.	The need to develop the reusability of space launch vehicle stages to prevent the generation of space debris, as well as to assess impact areas on Earth and provide warnings to residential zones.
35. Sustainability of on-orbit operations and on-orbit manufacturing	It is necessary to monitor and potentially regulate on-orbit operations and manufacturing to secure long-term sustainability. This is a necessary challenge to address in order to prevent unnecessary misunderstandings and conflicts between nations as commercial on-orbit services are being launched. There is a question as to why there is a specific focus on manufacturing. Future missions which include in-orbit activities such as servicing or manufacturing should ensure they are sustainable and do not have long-term impacts on the space environment. For example, the operations should consider debris mitigation. On-orbit operational safety is addressed by individual operators, ensuring sustainability of operations in outer space require stricter adherence to the Guidelines; the onus of implementation though national mechanisms falls on the Member States. On-orbit manufacturing, where spacecraft parts are manufactured and assembled in space, is an emerging field with promising usages in the areas of on-orbit servicing, and there has been very limited demonstration of this kind. With its inherent appeal to revolutionize the space sector and boosting revenues, on-orbit manufacturing requires more careful consideration to address sustainability requirements. There is limited prior experience and knowledge regarding various operational dimensions; therefore, it is necessary to consider its regulatory, supervisory and development aspects.	Missions performing in-orbit missions such as servicing or manufacturing are implementing “applicable international and national space debris mitigation standards and/or guides” (see guideline B.8, para. 2).

Section B. Safety of space operations		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
36. Failure to reach an international agreement on mechanisms and standards required for the implementation of some Guidelines that require data-sharing or consultation	Space activities affect all nations and services provided worldwide; thus, an international agreement is a must. This challenge is important to discuss, but additional clarification from the proposing country is needed. Existing guidelines mention "appropriate communication and consultation mechanisms" and "international intergovernmental organizations"; an exercise needs to be conducted to identify gaps, and match relevant organizations, which are identified through the implementation practices, which may fill the respective needs and gaps. The mechanisms are arrived at based on the mutual agreement between two entities involved in data-sharing. The lack of a universally agreed mechanisms for data exchange leads to additional efforts having to be made to meet diverse requirements. The requirements also depend on the specific entities involved in the data-sharing. While this is definitely a challenge, it is important to clarify whether the data-sharing or consultations mentioned should be carried out on a voluntary basis. The roots of this challenge likely include: – A reluctance among some leading spacefaring countries to share transparent data – The dominance of a competitive/commercial and political atmosphere in collaborative and international interaction This challenge could be reframed as the "Identification of obstacles hindering the definition of an international agreement on mechanisms and standards required to implement key guidelines that require data-sharing or consultations among Member States".	Continued reporting of implementation examples which highlight international intergovernmental organizations which serve to support the data-sharing. As part of the implementation sharing perform a critical review to maps common approaches and gaps. A common platform for data-sharing and streamlined procedures for sharing a particular type of data should be arrived at based on a consensus. The mechanisms can draw inputs from the most prevalent practices. Increased collaboration within the Committee on the Peaceful Uses of Outer Space framework in establishing infrastructure and the requirements for implementing regulations. A web portal-based interface is preferable for operational data exchange in a secured manner. Consultative Committee for Space Data Systems (CCSDS) standards are already used by several space agencies, the same can be adopted.

Section C. International cooperation, capacity-building and awareness		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline C.1. Promote and facilitate international cooperation in support of the long-term sustainability of outer space activities		
37. The promotion of interactive and cooperative approaches among members to avoid	It is necessary to discuss a multi-dimensional cooperative relationship including public-private partnerships, in light of the competitive development of the private sector after the Guidelines were agreed.	Member States are encouraged to share their expertise and experience through a variety of international forums in order to advocate the importance of responsible behaviours in space that will support the long-term sustainability of outer space activities.

Section C. International cooperation, capacity-building and awareness		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
a competitive space environment	The Working Group may consider increasing the activities related to raising awareness of the importance of cooperation between space countries. Commercial interests should be taken more into consideration when considering international cooperation. With the emergence of an increasing number of commercial operators, the potential for overexploitation of outer space is a real concern as regards individual space actors because focusing on immediate, short-term gains is more likely to take precedence over the considerations related to the long-sustainability of outer space. At the current stage, no specific challenge could be identified with regard to guideline C.1, " Promote and facilitate international cooperation in support of the long-term sustainability of outer space activities". The Office for Outer Space Affairs could play an important role, as demonstrated in the past, to support States in need and provide necessary advice and support. The dominance of a competitive, monopolistic and unregulated control approach has overshadowed attention to collective interests, conflicting with long-term sustainability goals and principles. This trend has particularly negative effects in areas where mutual cooperation, joint operations, synergy, transparency, trust, the exchange of data, knowledge and technology and resource-sharing are needed. At this stage, it seems that the first step to addressing this challenge is for leading spacefaring nations to take the initiative in this direction, recognizing the urgent need to reduce threats and enhance freedom of action in space. This movement will, in turn, guide developing countries to follow suit. Additional discussion would be beneficial to understand the challenge in more detail.	Through bilateral and multilateral consultations, Member States should promote the peaceful use of space and coordinate regulations that support long-term sustainability goals and other international obligations. States could consider mentioning the Office for Outer Space Affairs explicitly in the Guidelines, in section C (International cooperation, capacity-building and awareness) and the Office's potential mission. Strengthen and expand international research and development programmes. Foster regional and international networking. Additional discussion would be beneficial to understand the challenge in more detail and develop appropriate recommendations. Despite the economic and military approach to space, the intensification of competition in space is manageable, and its severity can be reduced. Avoiding a purely commercial, service-oriented and competitive approach in proceeding collaborations can be rigorously taken into account. This perspective should be continually promoted. "International cooperation and extensive capacity-building" are recognized as the primary factors for the implementation of guidelines. If implemented, the guidelines will lead to a reduction in capability gaps, balanced growth in achieving the objectives of the guidelines, and assistance to emerging and developing space countries. There is a need to eliminate political, legal and export restrictions in bilateral and multilateral relations and collaboration, and to facilitate the exchange of knowledge and technology among countries while ensuring oversight to prevent their misuse for non-peaceful purposes. It is important that a broad cross-section of stakeholders is engaged in long-term sustainability activities; this includes public, commercial and academic representatives in addition to relevant foreign Governments. The European Union's Horizon Europe has a third-party component to foster and promote international cooperation (see https://commission.europa.eu/funding-tenders/find-funding/eu-funding-programmes/horizon-europe_en). International conferences such as the United Nations Conference on Sustainable Lunar Activities, the World Space Forums and the International Astronautical Congress. European Centre for Space Economy and Commerce provides a European platform for interdisciplinary discussions and research on space economy and commerce (see: https://ecseco.org).

Section C. International cooperation, capacity-building and awareness		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		Organize informal discussions in conjunction with the meetings of the Working Group on the Long-term Sustainability of Outer Space Activities to provide an overview of current space activities relevant for the long-term sustainability of outer space activities. The objective is to facilitate interaction while promoting international cooperation.
38. Conflict of interest at national level	Conflicting objectives: Countries may have different priorities in space exploration, which can lead to conflicts and challenges in aligning with sustainability goals at the international level. Economic and strategic interest: this can also have an impact considering national security concerns and economic interests, which can create tensions and hinder collaboration.	
39. Encourage engagement between nations through diplomacy to reach consensus on urgent sustainability matters	Issues related to space debris and space traffic management require immediate discussions and consensus.	
40. Development of dedicated tools and practices for enhanced information sharing	The provision of article XI of the Outer Space Treaty to share and disseminate information (nature, conduct, locations, results of their space activities) about space activities with the international community, complements the registration of space objects. It promotes international cooperation in the peaceful exploration and use of outer space. Sharing information on the nature, conduct, locations and results of activities in outer space will help to mitigate risks of accidents, foster coordination and enable cooperation, thereby ensuring the peaceful, safe and sustainable use of outer space for the benefit of all. At present there are no standardized ways, tools or practices for submitting information under article XI of the Outer Space Treaty. It would complement the objects registration information by allowing States that are not launching states registering space objects to share information about their activities more widely.	States shall use the existing mechanisms of the treaties, find the ways to make them operational and develop tools for information sharing under article XI, for uniform, streamlined and easily comparable submissions. Building an easily accessible and searchable repository of such information, displayed in a uniform and easily comparable manner.

Section D. Scientific and technical research and development		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline D.2 Investigate and consider new measures to manage the space debris population in the long term		
41. Need to enhance the management of space debris	There is a need to enhance the management of space debris by adding continuous monitoring and regular reporting mechanisms to assess the effectiveness of implemented measures, promoting incentives for developing new technologies and innovative practices to mitigate debris, and encouraging public-private partnerships to accelerate the adoption of these Technologies. These additions will ensure a more proactive, innovative and collaborative approach to managing space debris, addressing both current challenges and future needs while maintaining the sustainability of space activities. This challenge could be better framed as the "mitigation of space debris". Given the competitive development of private sector after the agreement of the Guidelines, it is necessary to strengthen measures to mitigate space debris. Include the assessment of current measures that have been implemented at national, regional and international levels. Stimulate the exchange of good practices would prove to be effective. Which incentives are being referred to in the proposed paragraph 7? It would be ideal to specify them.	Proposal for additional paragraphs to be added to the end of existing guideline D.2: "6. States and international intergovernmental organizations should establish continuous monitoring and regular reporting mechanisms to assess the effectiveness of implemented measures for managing space debris. "7. States and international intergovernmental organizations should promote incentives for the development of new technologies and innovative practices to help mitigate the creation of new space debris. "8. States and international intergovernmental organizations should encourage public-private partnerships to accelerate the adoption and implementation of new space debris mitigation technologies." An alternative proposal for additional text to be added at the end of the existing guideline D.2 is as follows: "6. States and international intergovernmental organizations should develop and implement ongoing monitoring and regular reporting mechanisms to access the appropriate means to enhance the management of space debris. "7. States and international intergovernmental organizations should encourage the development of new technologies and innovate practices to mitigate the generation of new space debris." Countries that generated the majority of space debris should assume greater responsibility for addressing and reducing debris and facilitate capacity-building and effective knowledge and experience transfer for developing countries. Member States are encouraged to become a member of IADC and foster cooperation with other space agencies in order to support the long-term sustainability of space activities; and to encourage IADC to expand its activities to include emerging spacefaring nations to support capacity-building.
42. Potentially detrimental effects of metallic materials in the Earth's atmosphere due to the re-entry of artificial space objects	Recent research could identify measurable quantities of exotic metal particles in the higher atmosphere beside the natural meteoric space dust. The metal particles likely originate from the re-entries of artificial space objects. Additional research is ongoing to study the chemical and physical processes triggered by the metallic injection and possible detrimental effects to the environment.	Promote research and exchange on results and mitigation strategies.

Section D. Scientific and technical research and development		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
43. Potentially detrimental effects from rocket launches on the environment and atmosphere	The ever-increasing number of spacecraft requires an increasing number of rocket launches to bring them into orbit. During ascent, the heavy motors of the rockets deposit particles in all layers of the atmosphere and could even contribute to the pollution of the surroundings of the launch sites. Research is ongoing to understand the effects of the various propellants on the environment and how to mitigate them.	Promote research and exchange on results and mitigation strategies.

Note: There was an overall comment submitted by a Working Group member in connection with table I. There are no problematic issues related to the implementation of the Guidelines. For detailed information on the implementation of the Guidelines in national legislation, see conference room paper A/AC.105/C.1/2023/CRP.26 of the sixtieth session of the Scientific and Technical Subcommittee, entitled "Analysis of the Guidelines for the Long-term Sustainability of Outer Space Activities, adopted as a result of the sixty-second session of the Committee on the Peaceful Uses of Outer Space".

II. Challenges related to capacity-building in the framework of the Guidelines for the Long-term Sustainability of Outer Space Activities

Section A. Policy and regulatory framework for space activities		
Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline A.5. Enhance the practice of registering space objects		
44. Timely registration of space objects	Timely registration of space objects can be challenging for countries taking their first steps in the space industry, as newly developed space legislation and policies in such countries are new and untested. Whole-of-government collaboration as well as timely engagement with relevant external stakeholders is important for timely and accurate registration of space objects. Timely registration enables better data quality within the Register of Objects Launched into Outer Space, maintained by the Office for Outer Space Affairs, and associated national registries. A State agrees that there is a need for discussions on capacity-building, including the sharing of best practices regarding timely registration procedures. Timely space object registration secures timely calculation of ephemerides which leads to timely conjunction analysis, collision avoidance and dark and quite skies application. There aren't any consequences for not registering timely at the moment. Need for more precision: the term "timely" is somehow vague.	Development of recommendations on national space legislation in the context of space object registration. Study and publication of a report that outlines existing challenges, presents case studies on delayed registration practices, and explores emerging issues. All space system operators should liaise with the entity responsible for registration to register their satellites as soon as possible. Ensure that all (public and private) national space objects are registered within Member States. Member States should develop and maintain national registration policies and publish a procedure for the registration of National Space Objects. Member States should maintain a registry of National space objects and provides the Office for Outer Space Affairs with registration information on a timely manner. Setting up a platform for data exchange. Continued reporting of implementation examples which highlight the approach to timely registration of space objects.

Section A. Policy and regulatory framework for space activities

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	To be more precise, the Convention on Registration of Objects Launched into Outer Space mentions "as soon as practicable".	Better awareness-raising among all space actors.
	Registration is a key commitment as part of the Outer Space Treaty. Timely registration is a key consideration to ensure there is an up-to-date list of space objects.	A FAQ section could work.
	The latency involved in registering space objects should be minimized. Often new entrants in the space sector do not register their space objects or provide the pertinent information late.	The development of a space object registration template has eased the registration process by clearly defining the required data scope and offering detailed guidance on how to complete it.
	"Timely" should be quantified, and national legislations should be taken into account in the definition of "timely registration".	The "Space law for new space actors" project is another capacity-building mechanism, and its expansion could help mitigate this challenge.
	The definition of "timely" must be further discussed.	A key component involves education and outreach to ensure an understanding of the obligation to report and register all satellites.
		Focused efforts by Member States to ensure all space actors belonging to the State register their space objects in a timely manner.
45. Developing processes to enhance registration practices		Consider reviewing national space legislation to support identification of opportunities to enhance the practice of registering space objects.
		Consider dialogue with other States to share learnings, experiences and methods associated with registration.
46. How to address ongoing complexities related to the status of launching States	The expanding nature of novel space operations and business models means that identifying who should register a satellite is becoming more complex.	The Office for Outer Space Affairs should continue to build expertise amongst the user community and registration experts (i.e. through workshops and outreach events) such as occurred under the Registration Project.
	It is necessary to improve the legal framework for cases. For example, a space object is registered in the launching State "A", but later it transferred to another State "B", or the real user State of the space object is different.	States and international intergovernmental organizations shall consider bilateral and multilateral coordination when authorizing space activities in order to identify involved actors and define their roles.
	Modern space activities can be described by multiple co-launching States and other associated actors that contribute to carrying out space activities.	A follow-up to the Office for Outer Space Affairs Expert Event on Registration would be interesting to further this dialogue.
	Many States have varying definitions of "launching State" (See the Office for Outer Space Affairs Expert Event on the Registration of Objects Launched into Outer Space, held in Vienna in May 2023) and the responsibilities that the launching State bears.	States that are encountering complex registration cases should be encouraged to be transparent with their decision-making practices and share experiences with the expert and user community.
	This should be considered with regard to private actors cooperation/engagement in national activities.	The Office for Outer Space Affairs Expert Event on Registration, held in Vienna in 2023, could be seen as a best practice.
	Additional discussion would be beneficial to understand the challenge in more detail.	The notion of launching States is related to the notion of responsibility, to be addressed within the Legal Subcommittee.
	First of all, the "ongoing complexities" must be fully identified and then address them.	

Section A. Policy and regulatory framework for space activities

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
47. Data accuracy and completeness of space object registration	Incomplete registrations: some space operators may not provide complete information about their objects, leading to gaps in the registry. Verification difficulties: verifying the accuracy of the provided data can be challenging, especially for new or non-traditional operators.	
48. Standardization issues	Lack of uniform standards: different countries and organizations may have varying standards for what information should be included and how it should be formatted, creating inconsistencies. Diverse reporting practices: operators might follow different practices for reporting, making it difficult to consolidate and standardize data.	

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline B.2. Improve accuracy of orbital data on space objects and enhance the practice and utility of sharing orbital information on space objects		
49. Trackability and manoeuvrability of CubeSats and nanosats	There are challenges inaccurately tracking CubeSats and nanosats. When injected in batches of identical satellites, it is especially difficult to distinguished one from another and uniquely identify each one. Such satellites often do not carry on-board propulsion and are not manoeuvrable, hence incapable of on-orbit collision avoidance and re-orbiting/de-orbiting required for post-mission disposal. Above 400 km, and in particular in the area of human-inhabited space, CubeSats and nanosats should be trackable (if not manoeuvrable) because low Earth orbit is getting more and more congested. Further discussion is needed especially on the issue of manoeuvrability. Precision is needed on the term “nanosat”. This is a challenge because of the large number and small size of CubeSats launched and to be launched. Necessary if above a certain altitude Even small satellites can pose a risk to other space objects in the event of a collision. Trackability and manoeuvrability are important for enhanced protection in low orbital areas. On the other hand, it is necessary to consider whether it is economically and technically feasible to stipulate the possession of propulsion systems for CubeSats and nanosats and whether it is even necessary to have propulsion systems for such satellites.	Member States should consider mitigation measures for the use of CubeSats above 400 km that do not have deorbiting or manoeuvring capability. CubeSats should be examined with consideration to the 25-year rule. In particular, deorbiting capabilities/plan would be desirable, in the context of the long-term sustainability of outer space activities, for CubeSats and other classes of satellites. States and intergovernmental organizations should consider mechanisms to improve the trackability of small satellites (e.g. passive sensors). Compulsory registration and tracking of CubeSats. The satellites should be large enough to be tracked. Satellites lacking trackability and manoeuvrability should not be deployed at higher altitudes, so that their post-mission orbital lifetime can be limited. Propulsion systems for small satellites help add versatility of mission while addressing the sustainability requirements. Possible implementation of passive trackers on board nanosats, such as retroreflectors for laser-ranging tracking. Continued reporting of implementation examples which highlight how increased trackability is achieved. Propulsion systems for small satellites help add versatility of mission while addressing the sustainability requirements. Enforce manoeuvrability above altitudes that decay in more than one year.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	Additional discussion would be beneficial to understand the challenge in more detail. Challenge is linked to guideline B.8, paragraph 1, which discusses “approaches that increase the trackability of space objects”. Therefore, implementation examples are likely needed to review approaches. Specifically, more clarity is needed to understand if this challenge should be a change related to capacity-building (table II) or rather as a challenge related to implementation of the Guidelines (table I). It important to discuss tracking CubeSats in orbit, but additional clarification is needed. The trackability and manoeuvrability of CubeSats and nanosatellites have become significant challenges among some leading spacefaring nations with the rise in competition and the number of launches of these satellites. The exponential increase in satellites deployed may result in safe launch risks for other countries.	The primary responsibility for addressing this challenge lies with the leading spacefaring countries, which conduct the majority of CubeSat and nanosatellite launches. It is therefore recommended that data on these satellites be shared online for access by all countries, reducing launch risks for other nations and, through enhanced manoeuvrability, decreasing launch window constraints. Nonetheless, making the related technology affordable and accessible for all countries – especially developing nations (such as compact, high-endurance propulsion systems) – is essential. Consider cost-effective ways to deorbit as part of mission design and disposal. active or passive means to aid tracking and uniquely identify them may be installed on satellites. It is desirable to have on-board orbit determination capabilities aided by global navigational satellite systems so that a more accurate estimate of the orbit is available, and such orbital information can be shared by the small satellite operators for spaceflight safety.
50. Prevention of on-orbit failures of space systems, especially those that are mass-produced using commercial off-the-shelf components	Commercial off-the-shelf components offer a low cost and easily affordable alternative, but may not be space-qualified and hence more prone to on-orbit failures. Such a failure not only results in premature mission termination that renders an operational asset into a debris object but also may add to the risks of in-orbit break-up. In-orbit failures would increase space debris and increase threats to space assets. As it will be difficult to totally prevent failures (requesting better ground testing or systems reliability are solutions to lower the failure rate, but failures would still occur on mass-produced systems), we should also strive to reduce the consequences of these failures for the space environment. This challenge could be reframed as “improving reliability of post-mission disposal for large scale systems”. Additional discussion would be beneficial to understand the challenge in more detail. It is assumed that this challenge is related to the perceived lower reliability of commercial off-the-shelf components.	Reliability aspects should be addressed in the mission design. Required failure mode analysis should be conducted and adequate redundancy should be incorporated to impart better failure tolerance. A deorbiting mechanism should be included in any satellite to be launched. Enforce manoeuvrability above altitudes that decay in more than one year. For large constellations, it is encouraged to perform system-testing in an intermediate orbit, at least on the subsystems of the platform needed for decommissioning. It is also recommended to improve the reliability of post-mission disposal for large-scale systems, taking into account the number of satellites. Continued reporting of implementation examples which highlight to designing and testing systems using commercial off-the-shelf components. Reducing the cost of components with space-rating standards (instead of using commercial off-the-shelf components). Also, establishing a global platform for the supply of and access to space-grade components for all satellite-manufacturing countries. The satellites with such components can be tested at lower altitudes, and once the maturity of the technology is proven, it can be deployed at higher orbits. Before a satellite in a megaconstellation reaches its operational orbit, health tests must be carried out, in an intermediate orbit, on the subsystems of its platform required for withdrawal from service. For satellites operating in low-Earth orbit, this intermediate orbit must allow natural re-entry in less

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		than five years and have its apogee lower than the perigee of the operational orbit. Each satellite in a constellation must have a probability (P) of success for decommissioning operations (including passivation operations as well as decommissioning manoeuvres) according to the following rule: – Constellation with less than 50 satellites: P > 0.9 + N x 0.001 – Constellation with 50 or more satellites: P<0.95 – N corresponds to the number of satellites in the constellation.
51. Managing orbital congestion	As the number of satellites and missions increases, certain orbital regions are becoming crowded. This congestion complicates trajectory planning and increases the risk of collisions.	
Guideline B.6. Share operational space weather data and forecasts and **Guideline B.7. Develop space weather models and tools and collect established practices on the mitigation of space weather effects**		
52. Access to the necessary data, absence of an international database and limitations in knowledge and technology	The limited availability of data and the absence of an international database make it hard to share information and work together. Many countries lack the expertise needed in space weather, which affects the creation of accurate models and tools. Additionally, differences in technology levels among countries lead to gaps in capabilities, making it challenging to apply best practices for dealing with space weather effects.	
Guideline B.8. Design and operation of space objects regardless of their physical and operational characteristics		
53. Approaches to the design and operation of small-size space objects	Small-satellite platforms offer a very affordable and convenient means for embarking on a space venture and carry out technology demonstration. However, their size constraint often results in a lack of manoeuvrability and trackability, which has detrimental effects on the safety and sustainability of operating in outer space. Furthermore, such satellites are often launched as part of a constellation whose design emphasizes overall resilience of the system rather than addressing individual failures. As a result, the reliability requirements are likely to be relaxed for such smaller satellites and additional spare satellites are likely to be kept for quick replacements, both resulting in added space object population. With new technologies and technology demonstrations, some of the space objects lack orbit controlling systems. Educate the next generation of space engineers.	The requirements for trackability and manoeuvrability should be addressed at the design phase. The size should be large enough to enable tracking by conventional ground-based sensors. The small satellites should be preferably operated away from the overly crowded and highly utilized orbital regions, wherever feasibility exists, to minimize the requirements for on-orbit collision avoidance. The satellites should be equipped with satellite navigation-aided orbit determination capabilities, and the more accurate orbital information should be shared for the purpose of spaceflight safety. The contact information for the team dealing with on-orbit collision avoidance should be made available for coordination purpose. The defined approaches should take into account that many developing countries are entering the space industry through more accessible small-size space object missions to build their national capabilities. The approaches

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	Further discussion is needed on the extent to which this needs to be examined from a safety perspective. If the degree of freedom in satellite design is restricted, this may affect the realization of optimal design, which in turn may affect the development of new satellite technology, etc., by private companies, research institutes, etc. It is unclear what is the intention behind this. It is suggested not to make any distinction with regard to the size of the object, although some requirements are more difficult for some objects to achieve (e.g. identification of the object). This challenge is important to discuss, but additional clarification is needed. Additional discussion would be beneficial to understand the challenge in more detail.	should avoid negatively impacting or limiting the accessibility to such missions but instead offer guidelines that facilitate that better navigation is included as part of mission design and in the implementation stages of a mission. Guideline(s) should be created, and deorbiting technologies should be shared. Organize exchanges of information, training sessions on best practices for students and young graduates. States should identify where they need support so this can be provided by the Office for Outer Space Affairs and other members. Better practices on failure mode analysis can be adopted to prevent on-orbit failures. There should be adequate testing of the technology implemented on board a CubeSat or nanosat on the ground (pre-launch) or on a non-separable platform (such as the Polar Synchronous Launch Vehicle (PSLV) Orbital Experimental Module) to ensure technological maturity and robustness.

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
Guideline C.2. Share experience related to the long-term sustainability of outer space activities and develop new procedures, as appropriate, for information exchange		
54. Time and resources involved in setting up new procedures for information exchange	Information-sharing procedures and forums provide an opportunity for States to share information and insights. However, setting up procedures or forums for information exchange involves time and resources to, for example, convene regular meetings, draft reports or meeting papers, etc.	Encourage Member States to join forums for information-sharing exchange. Information-sharing forums to consider proactively inviting Member States to join.
Guideline C.3. Promote and support capacity-building		
55. Lack of required data, information, knowledge, technology and infrastructure to implement the Guidelines	Nations taking their first steps into the space industry, or considering entry, may find it challenging to develop policies and legislation that align with the recommendations of the Guidelines. This could be a challenge for emerging spacefaring nations, as the lack of knowledge concerning how to obtain the data and technology and a lack of infrastructure could hinder the implementation of the Guidelines. This lack is very much experienced by smaller States with smaller institutions that might not possess all the competencies required for the guideline's implementation. Besides, many guidelines can only be implemented when involving private industry.	Developing guidelines and standards that clearly define the scope of recommendations and establish commonly agreed approaches would assist nations in finding suitable solutions that align with their specific legislative frameworks. Member States are encouraged to conduct outreach with industry and academia to promote the development and use of techniques and methods to improve knowledge. This includes participating in conferences, workshops, and exercises. States and international intergovernmental organizations should consider consultations of the private industry either at a national level or though the

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	Any interference with the rights of space operators through legal measures should be proportional, appropriate and the least intrusive possible. Setting measures (legal or non-legal) at the national level requires contextual information in order to respect the principle of proportionality. For this assessment, data and knowledge are needed. To effectively implement the guidelines, it is essential to share the necessary and accurate data, information, knowledge and core technologies regarding what is going on in the orbits around the Earth. To ensure that emerging spacefaring nations enjoy inclusive participation in space activities, it is essential to develop frameworks that facilitate collaboration, knowledge-sharing and capacity-building initiatives. Establishing fair and transparent mechanisms for equitable access to low-Earth orbit is crucial, taking into consideration the needs and capabilities of all countries. Addressing gaps in data, information, knowledge, technology and infrastructure is vital for implementing the guidelines. This requires cooperation among all Member States, with strong support received by developing countries from developed nations to share expertise and resources.	Committee on the Peaceful Uses of Outer Space. Private industry shall be considered and consulted by the Working Group as well. The Office for Outer Space Affairs or other relevant bodies (e.g. European Space Policy Institute (ESPI) https://www.espi.or.at/) to support the exchange of information, via conferences, web tools or training tools. Members should identify where they need support so it can be provided by the Office for Outer Space Affairs/other members. Continued reporting on implementation practices is aimed at supporting capacity-building. Inclusion of such information in a repository will allow for ease of access to reporting practices. A comprehensive and integrated online data management system, including space situational awareness data, collision prediction, ephemerids, etc., must be established and be accessible to all space actors. Beyond just government-to-government interaction, space actors should look to industry and academia to develop knowledge, infrastructure and access to data. The role and potential contribution from the private industry and the Office for Outer Space Affairs shall be considered.
56. Mechanisms to ensure that emerging spacefaring nations enjoy inclusive participation in space activities	The infrastructure required for space science development is resource-intensive, and its management requires specialized expertise, which can be challenging for emerging spacefaring nations. This challenge is important to discuss, but additional clarification from the proposing country is needed. Any cooperation on space should be on a mutually acceptable basis. It should be considered the engagement with emerging spacefaring nations through dedicated initiatives in order to strengthen the cooperation with established space actors. A glossary and FAQ could help in this sense. The obstacles to the participation of all countries, particularly emerging space nations, hinders the advancement of the implementation of the guidelines. Given the orbital and frequency bandwidth congestion caused by the deployment of satellites within large constellations, equitable access to placing satellites in orbit has become a challenge for emerging countries. Since space is considered a common heritage of humanity, the benefits derived from space exploration should encompass all countries.	Launch capacity-building initiatives tailored for the public sector and non-profit organizations to raise awareness about the importance and potential positive impacts of the space sector, as well as its development challenges. Expand existing support mechanisms to enhance their reach. The enhancement of technology in emerging space countries through technology transfer and planning large collaborative projects involving all emerging space nations in the areas of satellite construction, space exploration, space situational awareness, space weather, space debris mitigation, and in-orbit servicing and operations, as well as equitable sharing of benefits, is definitely essential. There is a need to coordinate low-Earth orbit resources and to manage the status of large constellations to ensure equitable access for all countries and to establish a framework for the right allocation of these resources among all nations. Considering the importance of the principle of international cooperation and extensive capacity-building in the implementation of guidelines for the development of space exploration and the reduction of inequality in

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		capabilities, it is necessary to establish a comprehensive legal framework that encompasses the interests of all countries. Encourage Member States to join forums for information sharing. Information sharing forums to consider proactively inviting Member States to join. The Access to Space for All initiative is a successful example of providing developing nations with opportunities to access modern space infrastructure and broaden their network of cooperation. Regional centres enhance access to space education and support the development of space capabilities.
57. Mechanisms to ensure equitable access to low-Earth orbit	The equitable, rational and efficient use of the various orbital regions, including radio frequency and geostationary orbits as well as low-Earth orbit, is already included in the existing guideline A.4 – Ensure the equitable, rational and efficient use of the radio frequency spectrum and the various orbital regions used by satellites. It is desirable to discuss equitable access to low-Earth orbit under guideline A.4. Any cooperation on space should be on a mutually acceptable basis. The obstacles to the participation of all countries, particularly emerging space nations, hinders the advancement of the implementation of the guidelines. Given the orbital and radio frequency spectrum congestion caused by the deployment of satellites within large constellations, equitable access to placing satellites in orbit has become a challenge for emerging countries. Since space is considered a common heritage of humanity, the benefits derived from space exploration should encompass all countries.	The enhancement of technology in emerging space countries through technology transfer and planning large collaborative projects involving all emerging space nations in the areas of satellite construction, space exploration, space situational awareness, space weather, space debris mitigation and in-orbit servicing and operations, as well as equitable sharing of benefits, is definitely essential. There is a need to coordinate low-Earth orbit resources and to manage the status of large constellations to ensure equitable access for all countries and to establish a framework for the right allocation of these resources among all nations. Considering the importance of the principle of international cooperation and extensive capacity-building in the implementation of guidelines for the development of space exploration and the reduction of inequality in capabilities, it is necessary to establish a comprehensive legal framework that encompasses the interests of all countries.
58. Mechanisms to address the lack of required data, information, knowledge, technology and infrastructure to implement the Guidelines	Smaller and inexperienced operators often do not have in-house astrodynamics experts. Utilizing methods and software that are commonly accepted would ensure more reliable on-orbit ephemeris data are shared with space situational awareness service providers. Any cooperation on space should be on a mutually acceptable basis. The obstacles to participation of all countries, particularly emerging space nations, hinders the advancement of the implementation of the guidelines. Given the orbital and frequency congestion caused by the deployment of satellites within large constellations, equitable access to placing satellites in orbit has become a challenge for emerging countries. Since space is considered a common heritage of humanity, the benefits derived from space exploration should reach all countries.	Sharing methods/software for calculating on-orbit ephemerides. Sharing good practices in informal meetings with emerging spacefaring nations. The enhancement of technology in emerging space countries through technology transfer and planning large collaborative projects involving all emerging space nations in the areas of satellite construction, space exploration, space situational awareness, space weather, space debris mitigation and in-orbit servicing and operations, as well as equitable sharing of benefits, is definitely essential. There is a need to coordinate low-Earth orbit resources and to manage the status of large constellations to ensure equitable access for all countries and

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		to establish a framework for the right allocation of these resources among all nations. Considering the importance of the principle of international cooperation and extensive capacity-building in the implementation of guidelines for the development of space exploration and the reduction of inequality in capabilities, it is necessary to establish a comprehensive legal framework that encompasses the interests of all countries.
59. Need for the cooperation of all Member States and the full support of developed countries in the implementation of research and the sustainability of space exploration	This challenge is important to discuss, but additional clarification is needed. Additional discussion would be beneficial to understand the challenge in more detail. At present, the stated challenge is too vague to address. This may not be a challenge but rather a natural development of the Working Group activities and support among Member States. The obstacles to participation of all countries, particularly emerging space nations, hinders the advancement of the implementation of the guidelines. Given the orbital/frequency congestion caused by the deployment of satellites within large constellations, equitable access to placing satellites in orbit has become a challenge for emerging countries. Since space is considered a common heritage of humanity, the benefits derived from space exploration should encompass all countries.	Promoting benefits and role of the Committee on the Peaceful Uses of Outer Space and attracting new States members. The enhancement of technology in emerging space countries through technology transfer and planning large collaborative projects involving all emerging space nations in the areas of satellite construction, space exploration, space situational awareness, space weather, space debris mitigation and in-orbit servicing and operations, as well as equitable sharing of benefits, is definitely essential. There is a need to coordinate low-Earth orbit resources and to manage the status of large constellations to ensure equitable access for all countries and to establish a framework for the right allocation of these resources among all nations. Considering the importance of the principle of international cooperation and extensive capacity-building in the implementation of guidelines for the development of space exploration and the reduction of inequality in capabilities, it is necessary to establish a comprehensive legal framework that encompasses the interests of all countries.
60. Overall capacity-building	This challenge is important to discuss, but additional clarification from the proposing country is needed. Additional discussion would be beneficial to understand the challenge in more detail. The current trend in the implementation of guidelines indicates a lack of progress in capacity-building strategies in emerging space countries, resulting in a development gap in the space sector between developed spacefaring nations and developing countries.	Permanent coordination of training courses and related events, along with the conducting space research projects. Equal and inclusive invitation of developing countries to participate in these programs and initiatives. Effective participation of developed countries in the transfer of knowledge and experiences, and providing training programmes without a commercial, competitive or political approach. Establishment and development of regional research-oriented and educational centres to provide modular training materials for the development of specialized project-based capacity-building. FAQ and factsheets can help with the development of capacity-building among emerging and established countries.

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
61. Resource constraints and infrastructure development	Limited financial and technical resources can impede the development of space capabilities, particularly for developing countries. Developing the necessary infrastructure for space operations, such as tracking systems and ground facilities, can be costly and complex.	
62. Mechanisms to facilitate knowledge transfer between spacefaring nations and emerging spacefaring nations		
Guideline C.4. Raise awareness of space activities		
63. Overall awareness-raising	This challenge is important to discuss, but additional clarification is needed. Additional discussion would be beneficial to understand the challenge in more detail. The implementation of guidelines is achievable only through awareness-raising and informing in areas such as clarifying the necessity of technology transfer to emerging space nations, refraining from imposing unilateral sanctions against the activities that are along with peaceful use of space by other countries, and addressing the impact of deploying large constellations on the saturation of orbital resources and restricting the space activities of developing nations.	Establishing more regional research-based centres in different potential countries. Overall information dissemination through the organization of training courses and side events to outline the current status of the space industry development. Coordinating on-site visits to relevant centres in developed countries. Awareness-raising to establish a technology transfer mechanism for emerging space countries.
64. Raising awareness of national regulations applicable to outer space activities	Improved education and awareness of national regulations applicable to outer space activities can support capacity-building and implementation of the Guidelines. Dedicated tools and mechanisms, as well as publicly available guidance, are ways of assisting the space sector to navigate the regulatory process, which may also reduce possible delays and costs.	Consider developing tools and mechanisms to support those navigating regulatory processes for outer space activities. Consider making available information on where to find existing guidance materials.

Note: A Working Group member made an overall comment in connection with table II: there are no problematic issues related to capacity-building under the Guidelines.

III. Identification and consideration of new challenges to the long-term sustainability of outer space activities

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
65. The safety of human space flight and space stations	Increase congestion in the low Earth orbit environment, including non-manoeuvrable objects and debris creation could pose a hazard to manned space stations. A State agrees on the necessity of consultations regarding guidelines for crewed spaceflight and low-Earth orbit safety, especially as the commercial use of low-Earth orbit space stations is beginning. No, it is not coherent with the principle of free access to space.	This challenge to be further discussed as an area for potential new guideline(s). Establishment of a database under the Committee on the Peaceful Uses of Outer Space framework for data-sharing and the requirement for knowledge transfer to conduct precise estimation of probable collision regarding space launches and satellites manned space systems. To be identified as a future area of work.
66. The supervision and safe conduct of nominal close-proximity rendezvous operations in the context of in-orbit servicing	Active debris removal, rendezvous and proximity operations require further consideration at the international level as it emerges as a key means to mitigate space debris and extend the life of satellites. Ensuring the safe conduct of rendezvous and proximity operations is important for the implementation of on-orbit services such as debris removal and satellite refuelling. Close-proximity and rendezvous operations is subjected to precise manoeuvres and in advance coordination that their safe conducting helps preventing collisions between spacecrafts. Rendezvous and proximity operations are topics that are covered as part of discussions in Geneva because they have an impact not only on safety of space operations, but also on space security, on the level of trust between space actors and the risk of potential misunderstandings and miscalculations. Discussions on these topics could take place during the Open-ended Working Group on PAROS in all its aspects. Moreover, it is unclear what is meant by supervision. This challenge could be reframed as “The safe conduct of nominal close proximity operations”. This challenge is important, and it is important to discuss rendezvous and proximity operations, but additional clarification is needed. Additional discussion would be beneficial to understand the challenge in more detail.	This should be further discussed as an area for potential new guideline(s) within the overall umbrella of “active debris removal”. Encourage States to conduct the rendezvous and proximity operations in a safe and transparent manner in accordance with procedures published by individual countries. The monitoring of close-proximity and rendezvous operations, in collaboration with the relevant country, and the online sharing of related space data, early warnings, and ensuring the owner consent for the execution of operations, all foster coordination among international organizations and involved countries. The necessity of obtaining permission from the beneficiary country for conducting “close-proximity rendezvous operations” assures the safety and optimal performance of space equipment and can extend the operational lifespan of spacecrafts. To be identified as a future area of work. A Government formulated “Guidelines on a license to operate a spacecraft performing on-orbit servicing” in 2021. Now, the Commercial Debris Removal Demonstration (CRD2 mission) has been conducted in accordance with the above guidelines, under which its objectives have been made publicly available.
67. Prevention of dangerous alterations of space environment parameters resulting	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space	Development of a set of additional guidelines aimed at addressing the identified objectives.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
from intentional modifications	– Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". This proposal was not accepted when it was initially proposed, nor in 2019, when the Guidelines for the Long-term Sustainability of Outer Space Activities were adopted. There has been no modification in the proposed wording. What specific issue(s) does this proposal intends to cover? The idea of "intentional" modifications relates to security issues that are tackled within Geneva formats. Need for further information on this challenge. Potential link to security topics, which would be outside the remit of the Committee on the Peaceful Uses of Outer Space.	The formulation of appropriate laws and regulations to prevent intentional actions in space that lead to negative changes in the space environment is highly necessary.
68. Implementation of policy aimed at precluding interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities	Encourage other States to develop guidelines and promote information-sharing (e.g. public attribution) related to unauthorized access, etc. Development of a set of additional guidelines aimed at addressing the identified objectives. Development of relevant technologies and promotion of safety measures on space systems to prevent unauthorized access (such as shielding, warning and monitoring systems, and software hardening) is recommended. A Government developed an "Action Plan for Cyber Security of Critical Infrastructure" in June 2022 as well as "Cyber Security Guidelines for Commercial Space Systems" in March 2023.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	– Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". Implementation of such policies is important because there are threat concerns such as cyberattacks on on-board hardware and software. This proposal was not accepted when it was initially proposed, nor in 2019, when the Guidelines for the Long-term Sustainability of Outer Space Activities were agreed. Moreover, it echoes to intentional threats that are covered within Geneva. Additional discussion would be beneficial to understand the challenge in more detail.	
69. Preclusion of activities that could damage foreign ground and information infrastructure related to space activities	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved	Development of a set of additional guidelines aimed at addressing the identified objectives.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". This proposal is similar to a previous one that was not considered appropriate. If the proposal is deemed to cover resilience of ground infrastructure, it is relevant to discuss it within the Committee on the Peaceful Uses of Outer Space. But if it is meant to address intentional threats, it belongs to Geneva discussions. Potential link to security topics which would be outside the remit of the Committee on the Peaceful Uses of Outer Space. The challenge could be reworded to: "Strengthening resilience of ground and information infrastructure related to space activities".	
70. Establishment of procedures and requirements for the safe conduct of operations resulting in the destruction of in-orbit space objects	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". Potential link to security topics which would be outside the remit of the Committee on the Peaceful Uses of Outer Space.	Development of a set of additional guidelines aimed at addressing the identified objectives.
71. Development of criteria and procedures for the active debris	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented):	Development of a set of additional guidelines aimed at addressing the identified objectives.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
removal and, in exceptional circumstances, for the intentional destruction of non-registered space objects	– Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled “Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities”.	
72. Appropriate solutions for the active removal and destruction of non-registered space objects	Active debris removal/rendezvous and proximity operations requires further consideration at the international level as it emerges as a key means to mitigate space debris and extend the life of satellites. This proposal was not judged appropriate when it was proposed, nor in 2019. It must be noted that there still is no consensus on the active removal and destruction of “unregistered” space objects.	That this challenge be further discussed as an area for potential new guideline(s). In case of the active debris removal, encourage the States to conduct the active debris removal transparently and safely in accordance with national legislation based on the published standards and procedures in each State. The necessity of defining a specific procedure for providing solutions for space debris removal includes the assessment of: 1. Ownership of the space object (regardless of whether it is registered or not). 2. The potential risk it poses to other objects. 3. The active or inactive status of the certain space object, aiming to minimizing the potential for damage to other space objects. To be identified as a future area of work. A Government formulated the “Guidelines on a license to operate a spacecraft performing on-orbit servicing” in 2021.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		Now, the Commercial Debris Removal Demonstration (CRD2 mission) has been conducted in accordance with the above guidelines, under which its objectives made been made publicly available.
73. Transparency and safety assurance to encourage private sector actors to implement space debris removal activities	Necessary to discuss regulation/oversight, in accordance with technical developments. Necessary to discuss among the international community best practices that could inform national regulatory practices. However, it is unclear what "transparency and safety assurance" is referring to. It is necessary to discuss regulation/oversight, in accordance with technical developments. It is necessary to have a comprehensive discussion rather than narrowing the focus solely to the private sector. Necessary to discuss regulation/oversight, in accordance with technical developments. All should remain prudent with this technology, which is not mature. It requires transparency and communication. States should rather work on avoiding the creation of debris. Additional discussion would be beneficial to understand the challenge in more detail. It is necessary to discuss regulation/oversight, in accordance with technical developments.	It may be useful to refer to the work of the IADC and other expert bodies that are considering space debris mitigation. Experts from industry should have the opportunity to contribute to working groups to discuss this challenge. Registration of missions through the State to ensure appropriate licences are provided. To be identified as a future area of work. The Japanese Aerospace Exploration Agency (JAXA) has developed RABBIT, a risk avoidance support tool based on debris proximity collision probability, which is available to the public.
74. Implementation of operational and technological measures of self-restraint on States' space activities in order to prevent adverse developments in outer space	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects – Appropriate solutions for active removal and destruction of unregistered space objects	Development of a set of additional guidelines aimed at addressing the identified objectives.

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
	A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". This proposal echoes intentional threats that are covered within Geneva. Additional discussion would be beneficial to understand the challenge in more detail. There is a potential link to security topics which would be outside the remit of the Committee on the Peaceful Uses of Outer Space.	
75. International/global coordination between existing and developing space situational awareness systems in the different regions of the world to support space safety and sustainability	International coordination will lead to higher redundancy of a service for data provision; it can decrease data latency and increase the space objects population coverage. A State agrees that this challenge related to space situational awareness is important to discuss, but additional clarification from the proposing country is needed. It is of primary importance to promote the exchange of information on space objects and orbital events between existing and developing space situational awareness systems in the different regions of the world in order to ensure the safety and sustainability of the increasing space traffic. This pragmatic approach makes it possible to avoid unnecessary duplications, and appears to be much more realistic than the establishment of a universal centralized space situational awareness system. One of the requirements for collision avoidance is the sharing of space situational awareness data between space situational awareness systems through a network-based framework for producing and distributing orbital data across different geographic regions of the world.	A joint platform could be established to unite open-source space situational awareness data, agreed recommendations on space situational awareness data specifications and the activity of the scientific community focused on space situational awareness research. It is recommended to promote communication, coordination and collaboration between existing or developing space situational awareness systems in the different regions of the world, through the routine exchange of information on space objects and events in order to improve space safety and sustainability. In practical terms, coordination between space situational awareness systems could translate into: – Sharing of a database of primary objects and user information, including spacecraft attributes and operational points of contact – Sharing of ephemerides and planned manoeuvres by spacecraft, provided that the latter agree – Sharing of information on risk assessment for a common list of satellites – Adoption of the same standard data formats and communication protocols, in order to cooperate effectively – Encouraging and facilitating compatibility and interoperability across the different space situational awareness systems. Looking at the long-term future, it could even go one step further in terms of a global approach for space situational awareness. Lessons learned from different existing initiatives could be considered, such as from: – The space weather advisories centres for International Air Navigation selected by ICAO or

Section B. Safety of space operations

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and best practices*
		– The International Committee on Global Navigation Satellite Systems (ICG) established in 2005 under the umbrella of the United Nations, which facilitates compatibility, interoperability, and transparency across the different civil GNSS services around the world (such as GPS, Galileo, GLONASS, Beidou and Gagan). Application of such best practices should be without prejudice to the national security, technological, economic or development interests of States. It is necessary to develop standards and templates for data exchange and sharing, based on the dissemination of multi-source information with the participation of all countries, to establish a database. A coordination mechanism should be established within the Committee on the Peaceful Uses of Outer Space and among Member States.
76. Effect of multiplication of space ports	This refers to the effect of the multiplication of launches on sustainability of space activities. This challenge is important, and it is important to discuss space ports, but additional clarification is needed. Additional discussion would be beneficial to understand the challenge in more detail.	

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
Guideline C.3. Promote and support capacity-building		
77. Assignment of ownership to space debris	The utility and the impact of such a measure shall be studied by the Working Group. This is an important issue to consider at the United Nations level. There are strong interlinkages to the work of the Legal Subcommittee which should be taken into consideration. There are many challenges from the legal and technical aspects, such as ownership, jurisdiction, and handling of technical information, etc. In order to ensure the safety and sustainability in outer space and facilitate on-orbit services such as active debris removal, it is necessary to further discuss to solve the above challenges. While large debris (spacecraft) can usually be identified by owner, there remains a question on how to tackle the ownership of small and/or older debris whose ownership may no longer be trackable. This is important to	Continued reporting of implementation examples which highlight good practice. Definition of debris versus space objects; authority responsible for determining ownership of debris; method for identifying small and untrackable debris.

Section C. International cooperation, capacity-building and awareness

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
	ensure that future operations are able to remove such debris to clear the space environment, without infringing on other Member States' operations. This challenge is important, and it is important to discuss space debris, but additional clarification is needed. The assignment of ownership to space debris, particularly given the high volume of fragments from space objects, presents significant technical and legal challenges. Technically, the sheer number of debris fragments complicates tracking and identification, making it difficult to ascertain ownership and responsibility for each piece. Legally, existing frameworks, such as the Outer Space Treaty, lack clarity on ownership rights over debris, leading to disputes between States and complicating liability for damages caused by these fragments.	

Section D: Scientific and technical research and development

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
78. Development and implementation of criteria and procedures for the preparation and conduct of space activities aimed at the active removal of space objects/debris from orbit	No, this is a point which should be naturally addressed by the industry and research institutions. Active debris removal/rendezvous and proximity operations requires further consideration at the international level as it emerges as a key means to mitigate space debris and extend the life of satellites. Active debris removal should be ensured in accordance with appropriate rules. This challenge is important, and it is important to discuss on active removal, but additional clarification is needed. It would be preferable to focus on best practices and international standards to allow for sufficient commercial competition whilst maintaining sufficient transparency.	This area should be further discussed as an area for potential new guideline(s) within the overall umbrella of "active debris removal." Active debris removal should be carried out transparently and safely, according to procedures published by individual countries. Development of a set of additional guidelines aimed at addressing the identified objectives. The following requirements are recommended: The executor, timing, and method of orbital removal operations, as well as the characteristics of the object subjected to removal, must be fully specified and known in advance. To be identified as a future area of work. A Government of formulated the "Guidelines on a license to operate a spacecraft performing on-orbit servicing" in 2021. Now, the Commercial Debris Removal Demonstration (CRD2 mission) has been conducted in accordance with the above guidelines, under which its objectives have been made publicly available.

Section D: Scientific and technical research and development

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
	The following regulatory functions should be implemented within the framework of the Integrated Space Operations Security Framework (other than those already implemented): – Implementation of self-restraint measures in outer space – Preclusion of interference with the operation of foreign space objects through unauthorized access to their on-board hardware and software – Prevention of modifications of the environment – Preclusion of activities that could impair foreign ground and information infrastructures related to space activities – Active removal – Safe conduct of operations for destruction of space objects appropriate solutions for active removal and destruction of unregistered space objects A working paper was submitted at the sixty-fifth session of the Committee on the Peaceful Uses of Outer Space (A/AC.105/2022/CRP.11), entitled "Considerations on key unresolved tasks of ensuring safety of space operations in the context of the long-term sustainability of outer space activities". The potential for damage to other close space objects during "active debris removal" operations is one of the major challenges in these types of operations.	
79. Good practices for active debris removal	Active debris removal/rendezvous and proximity operations requires further consideration at the international level as it emerges as a key means to mitigate space debris and extend the life of satellites. Formulation of good practices and lay a groundwork for requirements to be followed during active debris removal missions. However, this might be quite challenging at this moment when the technology and industry is just in their beginnings. As spacefaring nations are promoting legal frameworks and technology development for active debris removal, it is necessary to share best practices to maintain a sustainable orbital environment. Recommendations focused around technical good practice relating to how to conduct active missions, and focusing on the transparency of	That this area be further discussed as an area for potential new guideline(s) within the overall umbrella of "active debris removal." Further work in the framework of a Long-term Sustainability of Outer Space Activities Working Group focusing on the practices associated with transparency and encouraging safe and sustainable operations would be valuable. To ensure discussion captures new developments and capabilities industry should be part of the discussion. Potential to feature as an expert group in a future phase of long-term sustainability work. The Committee on the Peaceful Uses of Outer Space and countries actively engaged in debris removal are requested to share their successful procedures with other members. To be identified as a future area of work.

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
	missions and their planned targets ensures sufficient transparency and maintains a safe space environment. The potential for damage to other close space objects during "active debris removal" operations is one of the major challenges in these types of operations.	A Government formulated the "Guidelines on a license to operate a spacecraft performing on-orbit servicing" in 2021. Now, the Commercial Debris Removal Demonstration (CRD2 mission) has been conducted with the above guidelines, under which its objectives have been made publicly available.
80. Protection of the dark and quiet skies, including for astronomical observations	This is a challenge because of the increasing impact of large constellations of satellites on astronomy for the protection of the dark and quiet sky. Due to the high number of spacecraft sent in orbit (especially due to large constellations deployment), an important impact can be caused on astronomical optical observations from ground. It is crucial to discuss this challenge and that it be included in the Guidelines, not only for astronomy but also for ecosystems, and the protection of cultural heritage. The primary aim is to reduce light and radio pollution. The dark and quiet sky is essential for the advancement of our understanding of the Universe. The number of artificial objects in orbit around the Earth already severely limits astronomical observations not only in the optical domain but also in the radio domain. Moreover, they interfere the natural appearance of the night sky, which is an integral part of the natural and cultural heritage of all humankind. Without a sensible approach, the uncontrolled use of space may mean that terrestrial astronomy will be severely constrained in the near future, including the large survey telescopes already under construction, such as LSST or SKA, in which considerable public and private funding has already been invested. Satellite constellations are having unintended impacts on observing conditions for radio and optical astronomy. The connections held by First Nations peoples with the night sky may also be a relevant consideration in discussions on this challenge. The justification is well laid out in the conference room paper put forward by a number of co-sponsors that advocated for the creation of an agenda item on dark and quiet skies. This is an important factor to be considered. However, there is need to understand more. The light pollution caused by the reflection of light from satellite megaconstellations can disrupt astronomical observations and reduce the	To protect the dark and quiet sky, it is essential to adopt internationally mandatory rules for the use of outer space that mitigate the impact of artificial object and especially satellite megaconstellations on astronomical observations. It is encouraged to design, produce and operate each satellite forming a constellation in order to limit as much as possible the optical disturbance to astronomical observation from the ground or space. That this continues to be discussed within the Scientific and Technical Subcommittee with the advancement of best practices and possible recommendations brought to the subcommittee for consideration. Consider information arising from discussions under the single issue/item "Dark and quiet skies, astronomy and large constellations: addressing emerging issues and challenges", which was first included as an item at the sixty-second session of the Scientific and Technical Subcommittee, held in February 2025. Output should come from the consideration under the agreed agenda item of the Scientific and Technical Subcommittee of the Committee on the Peaceful Uses of Outer Space from 2025 onwards. Continued discussion is important in the Committee on the Peaceful Uses of Outer Space and ITU frameworks, as well as with the International Astronomical Union, on the impact of large constellations of satellites on astronomy for the protection of the dark and quiet sky. Development of technology to enhance optical magnitude and reduce signal reflection. Similar to as in ISO/TS 6434:2024, each satellite in a megaconstellation must be designed, produced and deployed with the aim of achieving an apparent magnitude of 7 or more, in order to limit optical disturbances for astronomical observations from the ground or space. To be identified within future work. Supporting multi-stakeholder discussions on developing practical solutions that seek to address unintended impacts of satellite constellations on astronomy.

Section D: Scientific and technical research and development

Challenges identified	*Why? Justifications and explanations*	*Possible recommendations and possible best practices*
	accuracy of collected data. By reducing light pollution, the quality of astronomical observations can be enhanced, increasing the likelihood of discovering new celestial bodies. Reducing artificial interferences, such as bright satellites, allows for the collection of more accurate scientific data, thereby contributing to advancements in the field of astronomy. This is particularly significant for studies of faint and distant objects.	
	There is a need for strict regulations regarding the reduction of light and signal pollution from large satellite constellations.	
81. Sustainability of deep space missions	There is a growing number of missions to the Moon and beyond in the coming decades and a need to ensure the long-term sustainability of these activities for future generations.	Extend the applicable existing Guidelines to deep space missions and consider what new challenges to sustainability will emerge in the context of these missions.
	Early consideration of sustainability will ensure long-term sustainability of other areas of space outside of Earth orbits.	This area would benefit from further work and discussion as a new challenge in relation to the long-term sustainability of outer space activities. There is potential to include it as part of the recommendations on future activities and work related to exploring and developing guidelines to address new challenges.
	As deep space exploration activities become more active, there is a need to consider conducting deep space exploration in a manner that takes into account the Policy on Planetary Protection developed by the Committee on Space Research (COSPAR), while protecting potentially life-supporting celestial bodies.	To be identified within future work.
	It is necessary to examine relative policy and legal regimes to identify controversial issues.	States to implement planetary protection considerations through their licensing procedures.